Extended results (report accompanying the code repository)

Identifiability-Aware Parameter Estimation and Frequency-Band Assessment of Equivalent-Circuit Models: Application to In Vivo Tissue Bioimpedance

International Journal of Mathematical, Engineering and Management Sciences

Syaiful Anam[1,*], Ahmad Zarkasi[2], Zahari Md Rodzi[3]

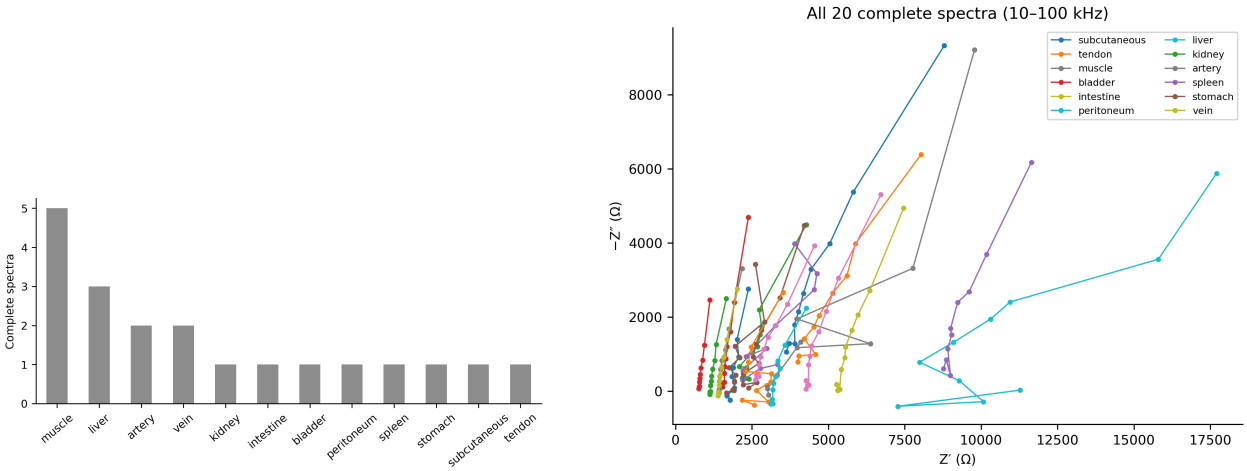
[1]Department of Mathematics, Faculty of Science, Technology, and Mathematics, Universitas Brawijaya, Malang 65145, East Java, Indonesia

[2]Department of Physics, Faculty of Mathematics and Natural Sciences, Universitas Mulawarman, Samarinda 75123, East Kalimantan, Indonesia

[3]School of Mathematics, Universiti Teknologi MARA (UiTM), Negeri Sembilan Branch, Seremban Campus, 70300 Seremban, Negeri Sembilan, Malaysia

[*]Corresponding author: Syaiful Anam, syaiful@ub.ac.id

Section, equation, table, figure and algorithm numbers without the prefix S refer to the main text. Notation follows the main text (Table 4 there); "calibrated" always means conditionally calibrated in the sense of Sect. 2.6.



(a) Complete spectra per tissue.

(b) Nyquist plots of all retained spectra.

Figure S1: EIS data retained after quality control (20 complete in vivo spectra, 10–100 kHz).

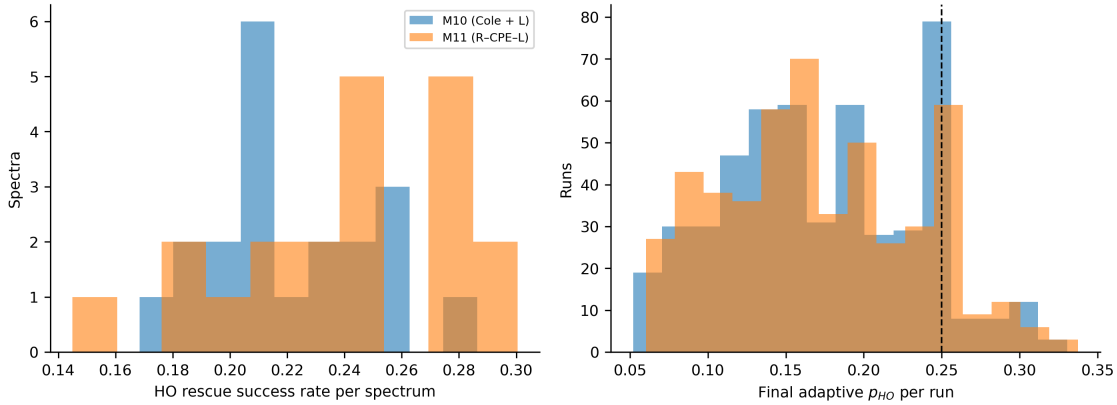


Figure S2: ASHO-LSHADE rescue behaviour: per-spectrum HO rescue success rate (left) and final adaptive p_{HO} per run (right; dashed line: initial value 0.25).

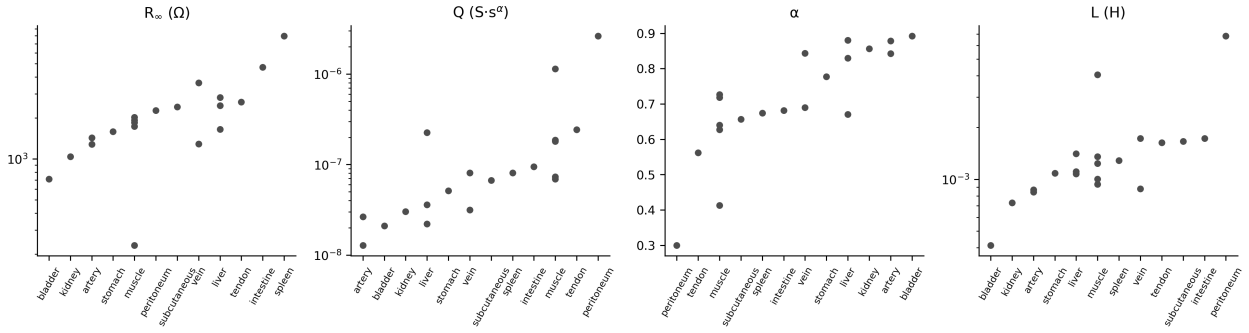


Figure S3: Identifiable M11 descriptors by tissue (one point per spectrum; descriptive only).

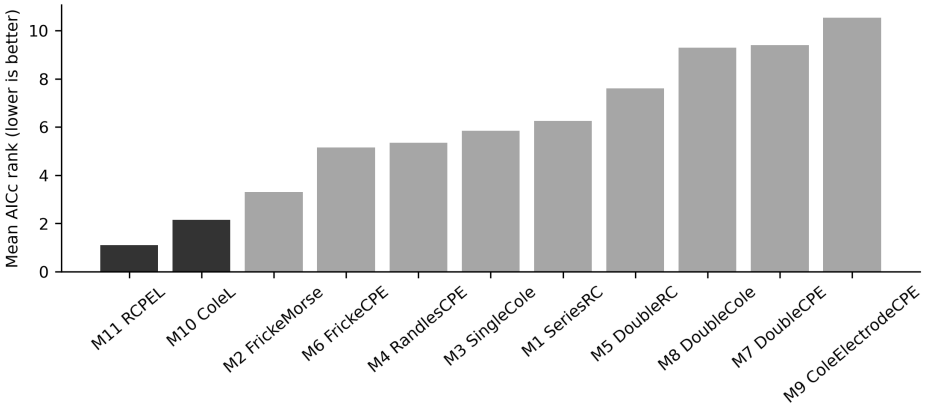


Figure S4: Mean weighted-AICc rank of the 11 candidate equivalent-circuit models (lower is better).

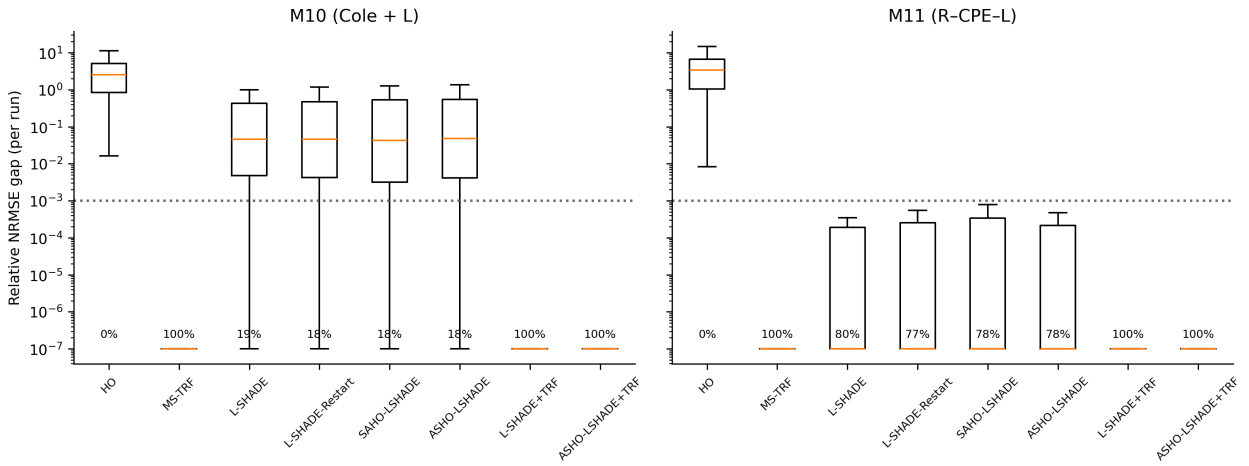


Figure S5: Run-level relative NRMSE gap to the best-known solution at 2,000 NFE; percentages give success rates; dotted line: 0.1% success threshold.

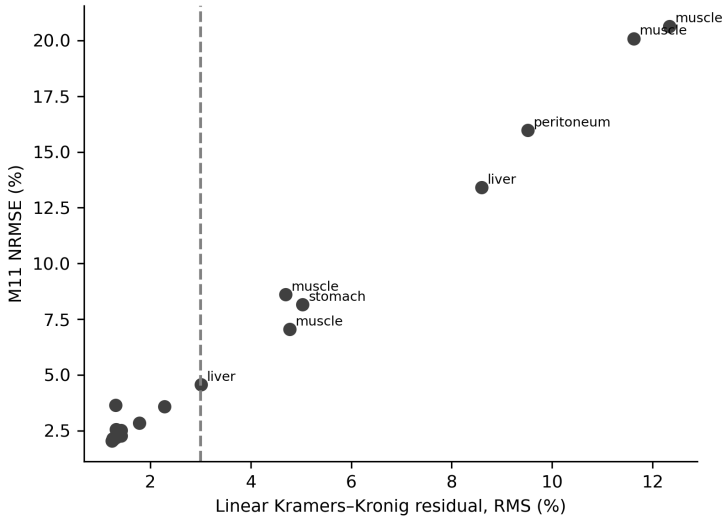


Figure S6: Linear Kramers–Kronig residual versus best-known M11 NRMSE; dashed line: 3% consistency threshold.

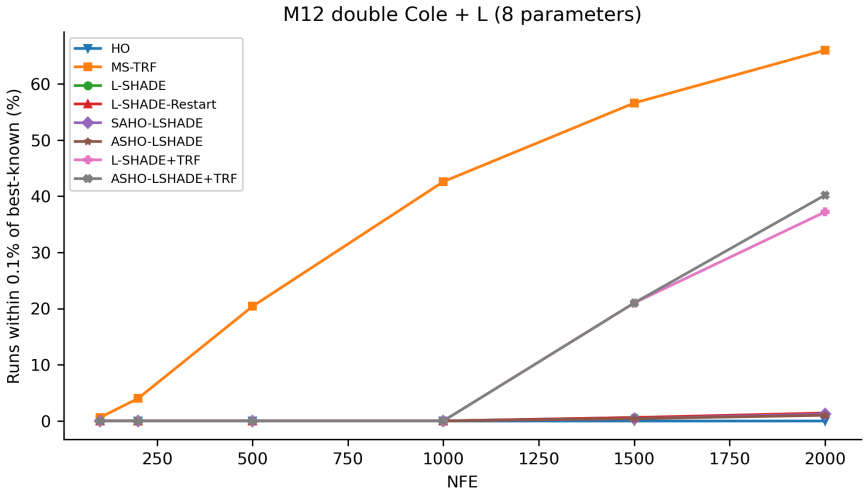


Figure S7: Higher-dimensional stress test (M12, eight parameters): percentage of runs within 0.1% of the best-known NRMSE versus NFE.

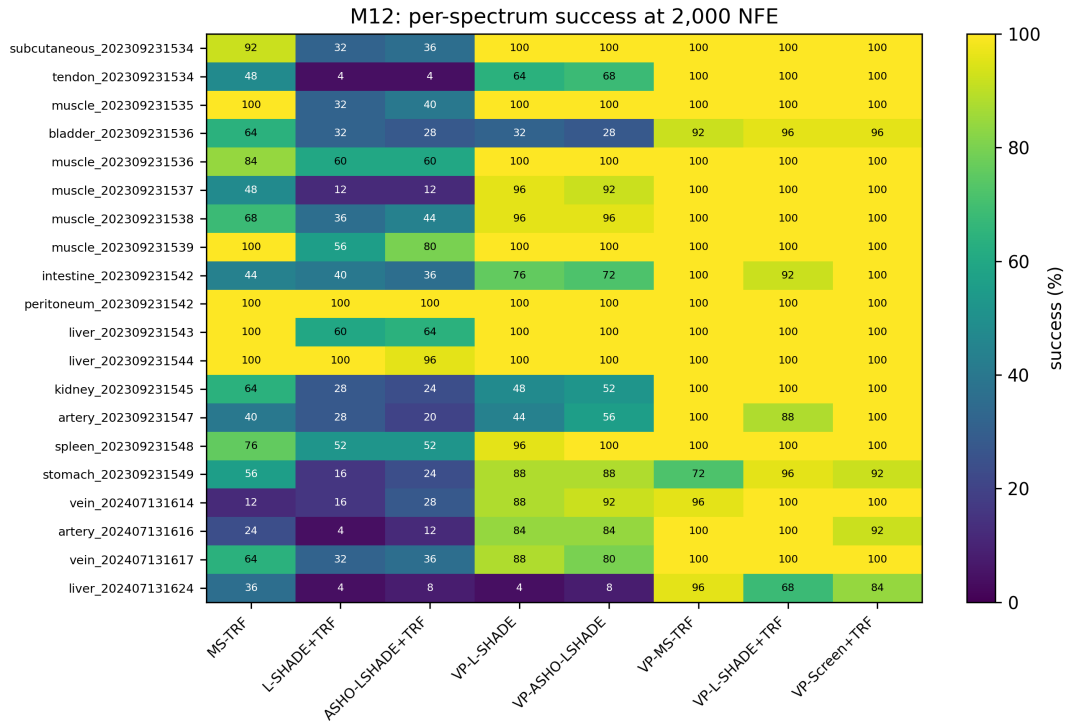


Figure S8: Per-spectrum success rate (%) of the eight-parameter M12 model at 2,000 NFE for the original multistart and memetic algorithms and the five variable-projection (VP) algorithms (25 runs per cell).

Table S1: Linear Kramers–Kronig (KK) residuals and best-known M11 NRMSE for the 20 complete spectra (KK-consistent: RMS residual $< 3\%$).

#	Tissue	Recording	KK RMS (%)	KK max (%)	KK-consistent	M11 NRMSE
1	subcutaneous	202309231534	1.31	4.28	yes	0.0364
2	tendon	202309231534	2.28	7.17	yes	0.0358
3	muscle	202309231535	4.69	10.50	no	0.0861
4	bladder	202309231536	1.27	4.68	yes	0.0209
5	muscle	202309231536	11.62	20.53	no	0.2009
6	muscle	202309231537	4.77	9.91	no	0.0705
7	muscle	202309231538	1.26	4.62	yes	0.0215
8	muscle	202309231539	12.33	32.76	no	0.2064
9	intestine	202309231542	1.42	5.10	yes	0.0226
10	peritoneum	202309231542	9.52	22.87	no	0.1597
11	liver	202309231543	3.01	5.86	no	0.0456
12	liver	202309231544	8.59	15.86	no	0.1340
13	kidney	202309231545	1.32	5.01	yes	0.0222
14	artery	202309231547	1.33	4.97	yes	0.0220
15	spleen	202309231548	1.41	4.81	yes	0.0228
16	stomach	202309231549	5.04	12.59	no	0.0816
17	vein	202407131614	1.42	5.17	yes	0.0250
18	artery	202407131616	1.32	5.06	yes	0.0255
19	vein	202407131617	1.24	4.88	yes	0.0204
20	liver	202407131624	1.79	6.31	yes	0.0284

Table S2: Frozen hyperparameters of the population-based algorithms.

Component	Setting
L-SHADE backbone	Initial population 20, minimum population 4 (linear reduction), memory size $H = 6$, $p = 0.11$, archive size equal to population size, $M_F = M_{CR} = 0.5$ initially
Rescue trigger	Stagnation $\geq \max(4, \mathrm{round}(12 - 7\,\mathrm{progress}))$ generations and normalized diversity $< 0.20 + 0.20\,\mathrm{progress}$; poorer half of the population eligible
SAHO-LSHADE	Fixed rescue probability 0.25
ASHO-LSHADE	$p_{\mathrm{HO}}^{(0)} = 0.25$, $p_{\min} = 0.05$, $p_{\max} = 0.45$, $\lambda = 0.25$
L-SHADE-Restart	Same trigger; uniform re-sampling with probability 0.25
HO	Population 20; position-update, defence and escape phases (Amiri et al., 2024)
Memetic variants	50% of NFE in the population stage, 50% in bounded TRF refinement

Table S3: Performance of all eight algorithms in the original parameterization across 20 spectra and 25 runs at 2,000 NFE. Success: NRMSE within 0.1% of the best-known value.

Algorithm	M10 (Cole + L)		M11 (R–CPE–L)	
	Success (%)	Median gap (%)	Success (%)	Median gap (%)
HO	0.0	254.3	0.0	343.2
L-SHADE	19.2	4.62	79.6	0.00
L-SHADE-Restart	17.8	4.57	77.2	0.00
SAHO-LSHADE	18.2	4.28	77.8	0.00
ASHO-LSHADE	18.4	4.85	78.0	0.00
MS-TRF	100.0	0.00	100.0	0.00
L-SHADE+TRF	100.0	0.00	100.0	0.00
ASHO-LSHADE+TRF	100.0	0.00	100.0	0.00

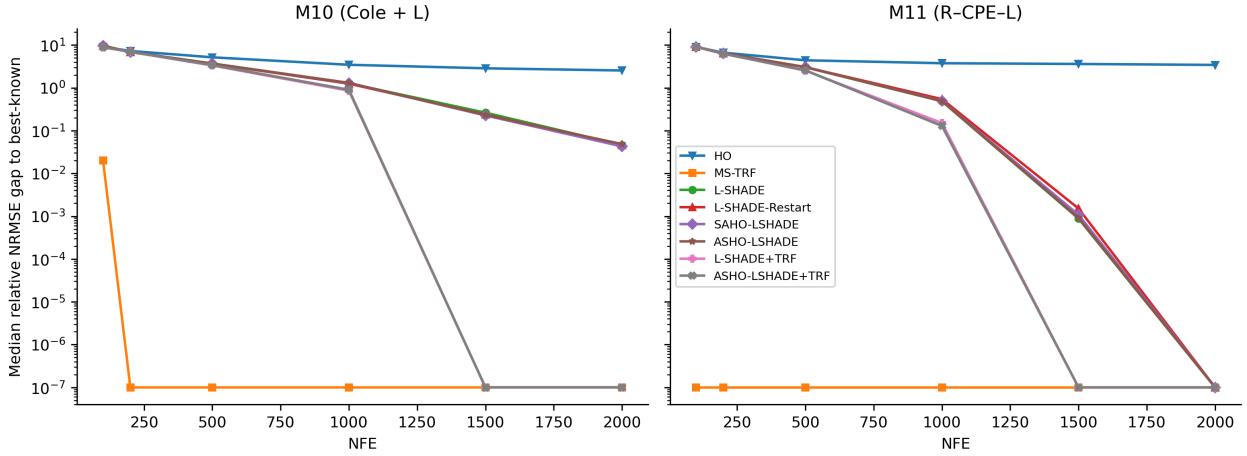


Figure S9: Original parameterization: median relative NRMSE gap to the best-known solution versus NFE (values below 10^{-7} are plotted at 10^{-7}).

Table S4: Sensitivity of the main conclusions to the sample: all spectra, leave-one-subset-out, and KK-consistent spectra only. M12 success at 2,000 NFE; gain: mean per-spectrum success difference VP-MS-TRF minus MS-TRF with bootstrap 95% CI over spectra (10,000 resamples); better/worse: spectra on which the spectrum-median NRMSE of VP-MS-TRF was lower/higher than that of MS-TRF by more than 0.1%; closed: spectra with a closed 95% profile interval (global VarPro profiles; closure only, without the width criterion of practical identifiability). The all-spectra interval comes from an independent bootstrap run over spectra; the interval used for inference (two-stage cluster bootstrap, 19.7–47.7) and the spectrum-resampling interval of the same run (20.4–43.6) are in Table S37.

	All spectra	Without subset A	Without subset D	KK-consistent
Spectra	20	4	16	12
M12 MS-TRF (%)	66.0	34.0	74.0	52.7
M12 VP-MS-TRF (%)	97.8	98.0	97.8	98.7
Gain, 95% CI (pp)	31.8 (20.6–43.6)	64.0 (46.0–80.0)	23.8 (13.5–34.5)	46.0 (33.7–58.3)
Better/worse	7/0	3/0	4/0	6/0
p VP-ASHO vs VP-L-SHADE	0.51	0.12	0.08	0.46
M10 $\log_{10}\tau$ closed	7/20	3/4	4/16	7/12
M10 $\log_{10}\Delta R$ closed	8/20	2/4	6/16	7/12
M11 $\log_{10} Q$ closed	20/20	4/4	16/16	12/12

Table S5: Weakly separable M13 model (Cole + L with a parallel CPE; 7 parameters, of which only L is linear): success rate (%) at increasing NFE and number of spectra (of 20) solved in all 25 runs. VP: partial variable projection eliminating L.

Algorithm	200	500	1,000	2,000	Spectra 25/25
L-SHADE	0.0	0.0	0.0	5.2	0
VP-L-SHADE	0.0	0.0	0.6	12.2	0
MS-TRF	33.2	86.2	98.6	99.8	19
VP-MS-TRF	52.4	93.8	99.4	100.0	20

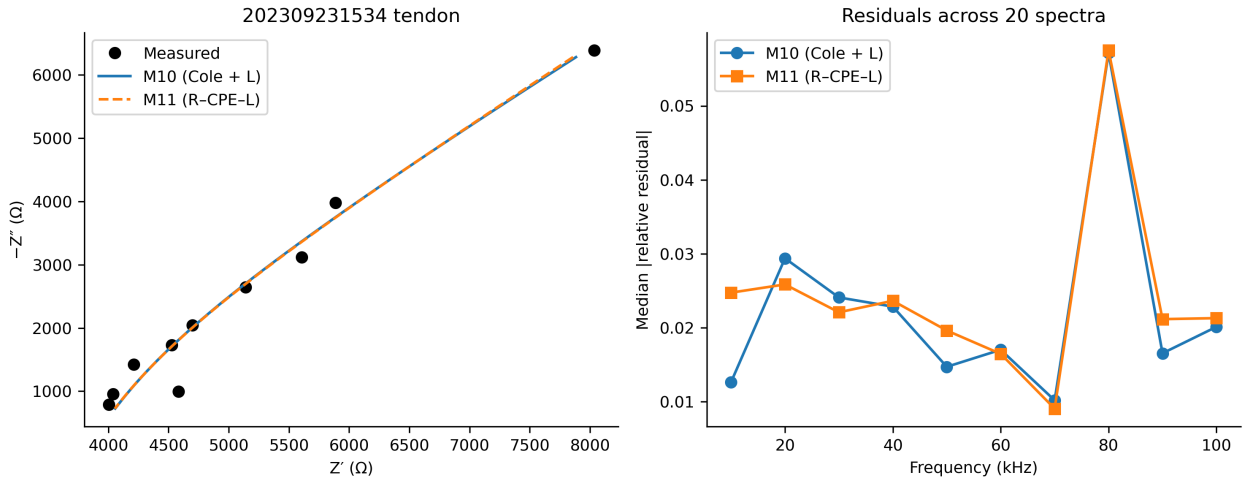


Figure S10: Representative M10 and M11 fits (left) and median absolute relative residual across all 20 spectra (right).

Table S6: Stability of fitted parameters as features for ranking spectra across 40 reading-level resamples: median and 5th percentile of the Spearman correlation between rankings from different resamples, and share of spectrum pairs ordered identically in at least 95% of resamples.

	KK-consistent (12 spectra, 66 pairs)			All (20 spectra, 190 pairs)		
Parameter	Median ρ	5th pct.	Stable pairs (%)	Median ρ	5th pct.	Stable pairs (%)
M10 $\log_{10}\tau$	0.50	-0.04	36	0.22	-0.22	13
M10 $\log_{10}\Delta R$	0.51	0.08	36	0.22	-0.17	13
M10 α	0.65	0.20	45	0.32	-0.11	16
M10 $\log_{10} R_\infty$	0.99	0.94	95	0.78	0.52	65
M11 $\log_{10} Q$	0.89	0.62	68	0.75	0.49	46
M11 α	0.87	0.64	71	0.80	0.53	53
M11 $\log_{10} R_\infty$	0.99	0.96	95	0.81	0.56	65

Table S7: Sensitivity of the Kramers–Kronig (KK) classification to the number of Voigt elements and to the threshold (20 spectra; reference: five Voigt elements, 3%).

Voigt elements	Median RMS (%)	Inconsistent spectra at threshold				Agreement with reference
		2%	3%	4%	5%	(of 20)
2	2.07	10	8	7	6	20
3	2.06	10	8	7	6	20
4	1.88	9	8	7	6	20
5	1.60	9	8	7	5	20
6	1.56	9	8	7	4	20
7	1.54	8	7	7	4	19

Table S8: Robustness of model selection (M10 Cole+L versus M11 R–CPE–L) and influence of the inductive term (20 spectra). Values are counts or median [interquartile range]. Leave-one-frequency-out here is pooled over all frequencies; Tables S18 (20 spectra) and S44 (12 KK-consistent spectra) split it into interior frequencies and band edges.

Quantity	Value
M11 preferred by AICc, weighted (relative) error scale	18 of 20
M11 preferred by AICc, constant error scale	18 of 20
Leave-one-frequency-out: M11 lower prediction error	14 of 20
Leave-one-frequency-out median relative error, M10 / M11	0.036 / 0.040 (Wilcoxon $p = 0.11$)
M10 solutions at bounds: α upper / $\log_{10}\Delta R$ upper / $\log_{10}\tau$	2 / 2 / 1
M11 solutions at bounds: α lower / upper	1 / 0
Effective inductance L (mH)	1.171 [0.920, 1.639]
ωL at 100 kHz relative to $\lvert Z\rvert$	0.311 [0.249, 0.381]
M11 NRMSE with / without inductive term	0.032 [0.023, 0.083] / 0.139 [0.099, 0.169]
$\lvert\Delta\log_{10} Q\rvert$ when 90–100 kHz excluded (decades)	0.051 [0.036, 0.072]
$\lvert\Delta\alpha\rvert$ when 90–100 kHz excluded	0.011 [0.007, 0.017]
$\lvert\Delta\log_{10} R_\infty\rvert$ when 90–100 kHz excluded (decades)	0.012 [0.006, 0.033]

Table S9: Equivalence analysis of the hippopotamus rescue: mean per-spectrum difference (rescue variant minus its counterpart) in success rate (percentage points) and in spectrum-median NRMSE (%), with 95% bootstrap confidence intervals over spectra. Equivalence margins: ± 5 percentage points and $\pm 1\%$. Cluster-aware intervals and two one-sided tests are given in Table S37.

Model	Comparison	Δ success (pp)	Within ±5	Δ NRMSE (%)	Within ±1%
M10	ASHO-LSHADE - L-SHADE	-0.8 [-3.6, 1.8]	yes	3.24 [-2.01, 12.08]	no
M10	VP-ASHO-LSHADE - VP-L-SHADE	0.0 [0.0, 0.0]	yes	0.00 [0.00, 0.00]	yes
M10	ASHO-LSHADE+TRF - L-SHADE+TRF	0.0 [0.0, 0.0]	yes	0.00 [0.00, 0.00]	yes
M11	ASHO-LSHADE - L-SHADE	-1.6 [-4.2, 1.0]	yes	0.00 [0.00, 0.01]	yes
M11	VP-ASHO-LSHADE - VP-L-SHADE	0.0 [0.0, 0.0]	yes	0.00 [0.00, 0.00]	yes
M11	ASHO-LSHADE+TRF - L-SHADE+TRF	0.0 [0.0, 0.0]	yes	0.00 [0.00, 0.00]	yes
M12	ASHO-LSHADE - L-SHADE	-0.2 [-1.2, 0.6]	yes	-0.82 [-4.89, 2.98]	no
M12	VP-ASHO-LSHADE - VP-L-SHADE	0.6 [-1.2, 2.4]	yes	-0.05 [-0.14, 0.00]	yes
M12	ASHO-LSHADE+TRF - L-SHADE+TRF	3.0 [0.0, 6.4]	no	-0.23 [-0.57, -0.02]	yes

Table S10: Sensitivity of the closure of the $\log_{10}\tau$ profile interval to the search bounds (20 spectra, nominal threshold): closure alone is bound-dependent, so identifiability also requires a width criterion.

Bounds of $\log_{10}\tau$	Closed (of 20)	Median width
[-6,0]	2	4.00
[-7,1]	7	4.98
[-9,3]	16	5.94

Table S11: Sensitivity of the benchmark: (a) success rate (%) at 2,000 NFE for alternative success tolerances; (b) L-SHADE success on M10 for different initial population sizes (20 spectra × 10 runs); (c) same-machine time to target of L-SHADE and VP-L-SHADE (20 spectra × 5 runs; unsuccessful runs counted as infinite).

(a) Success tolerance Model	Tolerance	L-SHADE	VP-L-SHADE	MS-TRF	VP-MS-TRF	VP-Screen+TRF
M10	1%	30.6	100.0	100.0	100.0	100.0
M10	0.1%	19.2	100.0	100.0	100.0	100.0
M10	0.01%	11.6	100.0	100.0	100.0	100.0
M12	1%	3.2	87.4	70.8	99.2	99.4
M12	0.1%	1.2	80.2	66.0	97.8	98.2
M12	0.01%	0.8	77.8	62.4	97.8	98.2

(b) Initial population size of L-SHADE (M10)		
10: 15.0%	20: 21.0%	40: 7.5%

(c) Time to target Model	Algorithm	Success (%)	Median time, all runs (ms)	Median NFE, successful
M10	L-SHADE	21	undefined (>50% failed)	1686
M10	VP-L-SHADE	100	22	238
M12	L-SHADE	1	undefined (>50% failed)	1557
M12	VP-L-SHADE	81	128	997

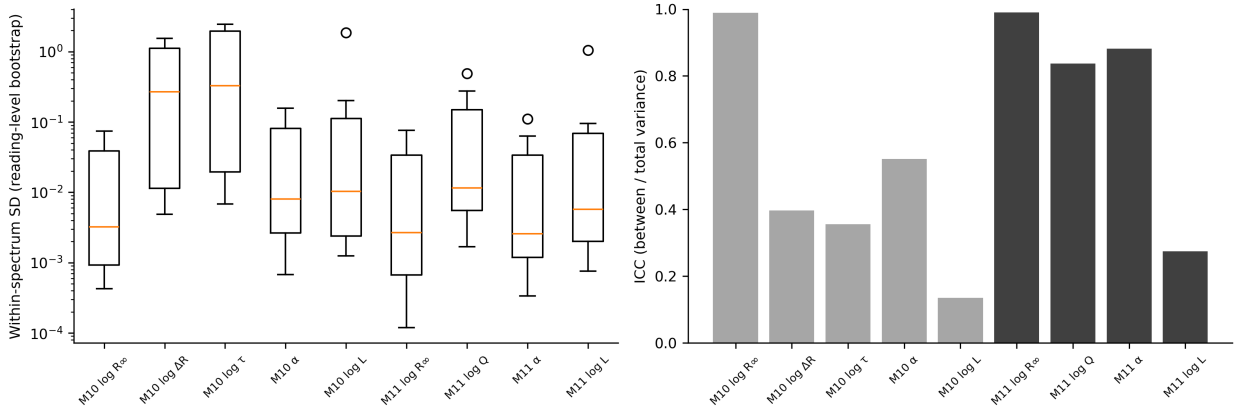


Figure S11: Measurement-level repeatability for the 12 KK-consistent spectra: within-spectrum SD of each parameter across 40 reading-level resamples (left) and intraclass correlation across spectra (right). Light bars: M10; dark bars: M11.

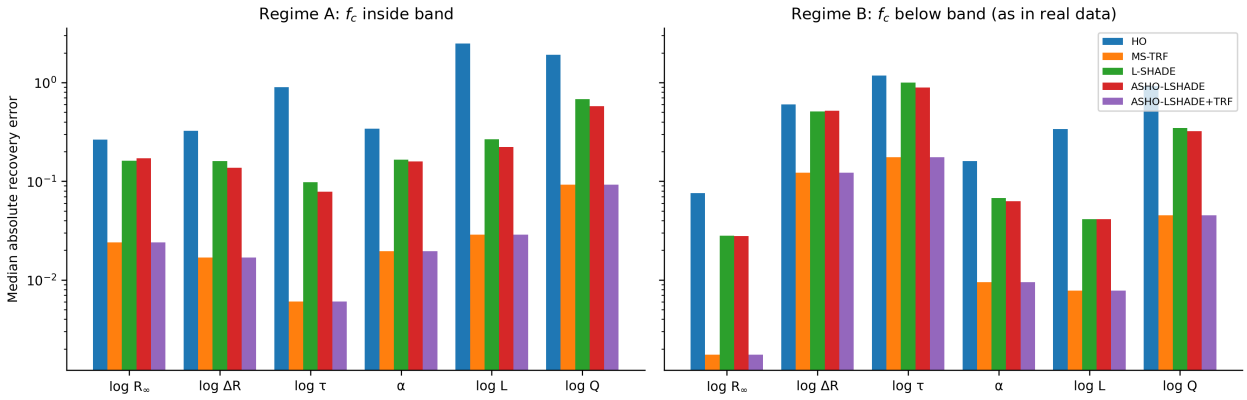


Figure S12: Median absolute parameter-recovery error on synthetic spectra with 1% noise. Regime A: f_c inside the measured band; regime B: f_c below the band, as in the measured tissue spectra.

Table S12: Identifiable M11 parameters across the 20 spectra (best-known solutions).

Parameter	Median	Q1	Q3	Minimum	Maximum
R_∞ (Ω)	1888	1396	2505	232	7978
Q (10^{-8} S s$^\alpha$)	7.12	3.13	18.2	1.28	264
α	0.704	0.652	0.843	0.300	0.892
L (mH)	1.17	0.92	1.64	0.41	6.81
NRMSE	0.0321	0.0225	0.0827	0.0204	0.2064

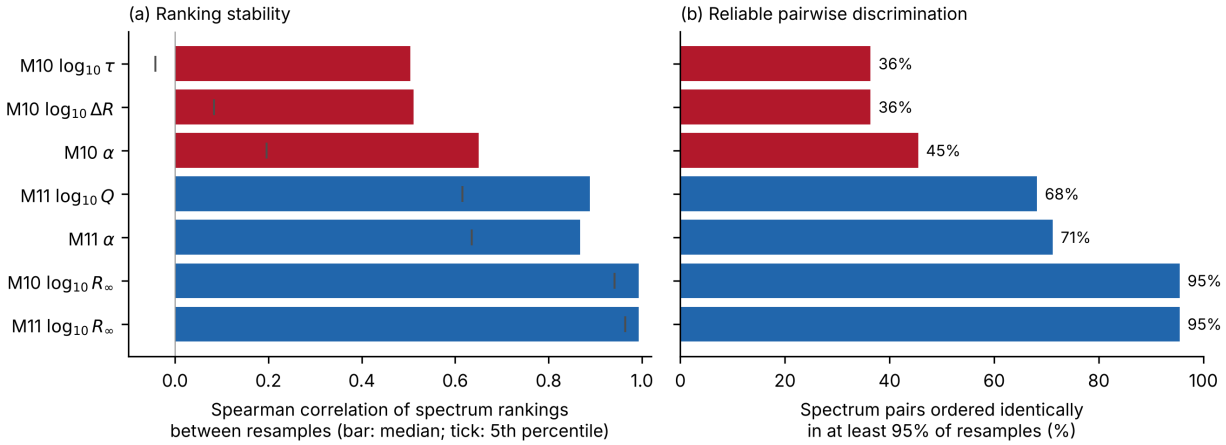


Figure S13: Stability of fitted parameters as features for ranking the 12 KK-consistent spectra across 40 reading-level resamples. (a) Spearman correlation between rankings from different resamples (bar: median; tick: 5th percentile). (b) Share of spectrum pairs ordered identically in at least 95% of resamples. Red: non-identifiable Cole parameters; blue: identifiable parameters. Reading-level resampling understates within-recording variability (Table S19).

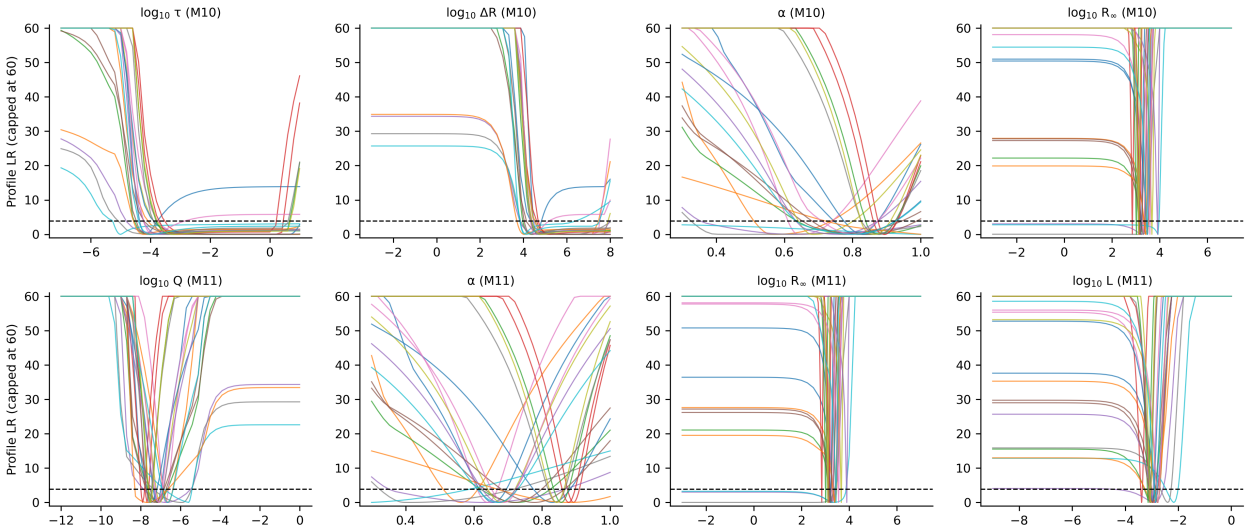


Figure S14: Nominal profile likelihood-ratio curves for all 20 spectra (one line per spectrum; dashed line: χ^2_1 threshold 3.84). Calibrated thresholds were 1.8–2 times larger (Table S14). Flat profiles for $\log_{10}\tau$ and $\log_{10}\Delta R$ in M10 indicate practical non-identifiability.

Table S13: Empirical error model (Eq. (5) of the main text) estimated from the R–CPE–L residuals of the 51 complete sweeps; 95% intervals from resampling recordings. Residual-based: residual vectors treated as errors (used for calibration). Projection-corrected: likelihood evaluated in the 16-dimensional residual space of each sweep, which accounts for the projection of errors by the fit; it is weakly identified, because a smooth error component that lies in the model tangent space is absorbed by the parameters and leaves no trace in the residuals (Table S23). [a]Mean RSS of the sweep fits divided by the mean squared deviation of sweeps from the recording mean (12 recordings with ≥ 2 sweeps).

Quantity	Residual-based estimate	95% interval	Projection-corrected	95% interval
σ_0 (relative scale, real part)	0.110	0.019 to 0.159	0.356	0.076 to 0.504
γ (frequency trend)	0.009	-0.212 to 0.233	0.021	-0.308 to 0.191
ρ_f (adjacent-frequency correlation)	0.180	-0.116 to 0.471	0.930	0.064 to 1.000
ρ_{RI} (real–imaginary correlation)	-0.194	-0.654 to 0.145	-0.370	-0.719 to 0.596
κ (imaginary/real scale)	0.459	0.288 to 1.574	0.473	0.280 to 3.620
Sweeps / recordings / residual df	51 / 20 / 816			
Sample lag-1 correlation, real / imaginary	0.37 / 0.16			
Ledoit–Wolf shrinkage intensity	0.91			
Share of sweep-to-sweep deviation left as residual[a]	0.15	IQR 0.07–0.59		

Table S14: Coverage (%) of 95% profile-likelihood intervals for the R–CPE–L descriptors (12 KK-consistent spectra $\times$ 100 replicates per condition; cluster-bootstrap 95% intervals over spectra for the empirical-Σ rows: nominal 78.8–86.0, calibrated 90.9–95.5, GLS 89.3–93.7, GLS calibrated 90.5–96.4). Nominal: χ^2_1 threshold 3.84; calibrated: parametric-bootstrap critical value (200 replicates) under the empirical Σ; GLS: generalized least squares with the empirical Σ. Misspecified: Cole data with f_c 200 Hz–3 kHz sharing the M11 high-frequency asymptote. [a]Critical values recomputed at each replicate's own estimate (25 replicates $\times$ 80 inner replicates per spectrum). [b]Cole truth with f_c below the band, empirical Σ, 50 replicates per spectrum. Median critical values: independent likelihood 6.8 ($\log_{10} Q$), 7.4 (α), 8.2 ($\log_{10} R_\infty$), 6.4 ($\log_{10} \tau$), 6.0 ($\log_{10} \Delta R$); GLS 4.9, 4.9, 4.7, 4.3, 4.2.

Condition	Parameter	Nominal	Calibrated	GLS	GLS calibrated
Empirical Σ	$\log_{10} Q$	83.8	93.2	91.6	93.2
	α	82.9	92.9	91.7	94.0
	$\log_{10} R_\infty$	81.1	94.0	91.2	93.3
$\rho_f - 0.2$	$\log_{10} Q$	90.1	96.8	94.2	95.6
	α	89.8	96.6	94.2	95.7
	$\log_{10} R_\infty$	87.5	96.9	95.5	96.8
$\rho_f + 0.2$	$\log_{10} Q$	77.0	88.2	88.3	90.8
	α	75.7	88.2	87.8	91.0
	$\log_{10} R_\infty$	71.0	88.2	84.2	87.0
White	$\log_{10} Q$	90.8	97.5	88.8	90.8
	α	90.8	97.8	89.2	91.5
	$\log_{10} R_\infty$	90.8	98.4	92.2	94.9
AR(1), $\rho = 0.6$	$\log_{10} Q$	69.5	82.8	76.0	78.7
	α	69.7	83.8	76.3	78.8
	$\log_{10} R_\infty$	67.3	84.8	78.2	81.4
Misspecified (Cole truth)	$\log_{10} Q$	9.9	17.3	9.8	10.8
	α	11.9	22.0	12.8	14.8
	$\log_{10} R_\infty$	31.9	52.2	40.3	43.7
Double bootstrap[a]	$\log_{10} Q$	83.7	92.0	89.7	92.7
	α	81.7	92.7	90.3	92.7
M10, Cole truth[b]	$\log_{10} \tau$	84.5	94.8	93.8	95.2

Table S15: Measured 95% profile intervals (20 spectra) by procedure: number closed within the bounds, number practically identifiable (all / 12 KK-consistent) at width thresholds of 0.5, 1.0 and 1.5 decades for $\log_{10}$ parameters (0.1, 0.2 and 0.3 for α), and median width. [a] $\alpha \in [0.2, 1]$ and $1/Q \in [10^{-2}, 10^{14}]$ (default $[0.3, 1]$ and $[1, 10^{12}]$), 1.0-decade threshold.

Parameter	Procedure	Closed	0.5 (0.1)	1.0 (0.2)	1.5 (0.3)	Median width
M10 $\log_{10}\tau$	Nominal	7	0 / 0	1 / 1	2 / 2	4.984
	Calibrated	6	0 / 0	0 / 0	1 / 1	5.313
	GLS	6	0 / 0	0 / 0	0 / 0	5.312
	GLS calibrated	5	0 / 0	0 / 0	0 / 0	5.406
M10 $\log_{10}\Delta R$	Nominal	8	1 / 1	2 / 2	2 / 2	3.613
	Calibrated	7	0 / 0	1 / 1	1 / 1	3.706
	GLS	4	0 / 0	0 / 0	0 / 0	3.790
	GLS calibrated	4	0 / 0	0 / 0	0 / 0	3.812
M11 $\log_{10} R_\infty$	Nominal	17	17 / 12	17 / 12	17 / 12	0.031
	Calibrated	17	17 / 12	17 / 12	17 / 12	0.075
	GLS	19	17 / 12	19 / 12	19 / 12	0.061
	GLS calibrated	19	17 / 12	19 / 12	19 / 12	0.069
M11 $\log_{10} Q$	Nominal	20	11 / 11	16 / 12	17 / 12	0.467
	Calibrated	20	8 / 8	13 / 12	16 / 12	0.621
	GLS	20	8 / 8	15 / 12	16 / 12	0.564
	GLS calibrated	20	7 / 7	15 / 12	16 / 12	0.661
M11 α	Nominal	18	10 / 10	15 / 12	16 / 12	0.100
	Calibrated	16	6 / 6	13 / 12	14 / 12	0.137
	GLS	17	7 / 7	15 / 12	16 / 12	0.123
	GLS calibrated	17	6 / 6	14 / 12	16 / 12	0.143
M11 $\log_{10} Q$, wider bounds[a]	Nominal			16 / 12		0.457
	Calibrated			13 / 12		0.621
M11 α, wider bounds[a]	Nominal			15 / 12		0.100
	Calibrated			13 / 12		0.137

Table S16: Bias of the R–CPE–L descriptors when the data are a noise-free Cole dispersion sharing the same high-frequency asymptote (12 KK-consistent spectra; median over spectra). Bias = fitted value minus asymptotic value.

f_c (Hz)	$f_{\min}/f_c$	$\log_{10} Q$	α	$\log_{10} R_\infty$
10	1000	0.028	-0.005	-0.001
30	333	0.059	-0.011	-0.002
100	100	0.133	-0.026	-0.006
300	33	0.279	-0.055	-0.012
1000	10	0.617	-0.121	-0.030
3000	3	1.216	-0.241	-0.090

Table S17: Sensitivity of the R–CPE–L descriptors to the treatment of the high-frequency correction (20 spectra; median [interquartile range]). Δ: absolute difference from the published fit. AICc best: number of spectra in which the treatment had the lowest AICc among full-band treatments.

Treatment	NRMSE	$\lvert\Delta\log_{10} Q\rvert$	$\lvert\Delta\alpha\rvert$	$\lvert\Delta\log_{10} R_\infty\rvert$	Correction parameter	AICc best
Series L (published), 10–100 kHz	0.032 [0.023, 0.083]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	L (mH) 1.17 [0.92, 1.64]	13
No correction, 10–80 kHz	0.127 [0.087, 0.142]	0.530 [0.453, 0.627]	0.124 [0.105, 0.144]	0.062 [0.057, 0.086]	–	–
Series L, 10–80 kHz	0.032 [0.022, 0.085]	0.051 [0.036, 0.072]	0.011 [0.007, 0.017]	0.012 [0.006, 0.033]	L (mH) 1.38 [0.96, 1.93]	–
Pure delay $e^{-j\omega t_d}$	0.033 [0.024, 0.082]	0.053 [0.020, 0.082]	0.010 [0.004, 0.017]	0.038 [0.021, 0.055]	t_d (μs) -0.55 [-0.61, -0.43]	3
Constant phase offset $e^{j\varphi_0}$	0.068 [0.049, 0.093]	0.728 [0.666, 0.874]	0.143 [0.127, 0.167]	0.022 [0.016, 0.043]	φ_0 (rad) 0.19 [0.16, 0.23]	2
Generalized term $K(j\omega)^\nu$	0.030 [0.022, 0.082]	0.082 [0.007, 0.353]	0.019 [0.000, 0.083]	0.051 [0.004, 0.160]	ν 0.92 [0.60, 0.99]	2

Table S18: Uncertainty-aware comparison of M11 (R–CPE–L) and M10 (Cole+L) (number of spectra out of 20 preferring M11 unless stated). Bootstrap LR: 100 parametric-bootstrap replicates from the fitted M11 with the empirical Σ. LOFO: leave-one-frequency-out relative prediction error; p: Wilcoxon signed-rank test.

Criterion	Result
AICc, independent likelihood: 10–100 / 10–80 / 20–100 kHz	18 / 19 / 19
AICc, GLS likelihood: 10–100 / 10–80 / 20–100 kHz	20 / 20 / 19
Bootstrap LR test rejects M11 ($p < 0.05$): independent / GLS	2 / 0 (median p 0.29 / 0.36)
LOFO, interior 20–90 kHz: M11 better; medians M11 vs M10	14; 0.0351 vs 0.0345 ($p = 0.11$)
LOFO, band edges 10 and 100 kHz: M11 better; medians M11 vs M10	9; 0.0376 vs 0.0351 ($p = 0.76$)

Table S19: Sweep-level repeatability (complete sweeps refitted by global VarPro; six KK-consistent and six KK-inconsistent recordings with ≥ 2 sweeps) compared with the earlier reading-level resampling (12 KK-consistent spectra, 40 resamples). SD: median within-recording standard deviation; ICC: one-way intraclass correlation over the KK-consistent recordings with 95% interval from 2,000 resamples of recordings; with six recordings the sweep ICCs are imprecise.

Parameter	Sweep SD KK-consistent	Sweep SD KK-inconsistent	Sweep ICC [95% CI]	Reading-level SD	Reading-level ICC
M10 $\log_{10}\tau$	0.958	0.374	0.48 [-0.16, 0.80]	0.329	0.36
M10 $\log_{10}\Delta R$	0.632	0.313	0.43 [-0.11, 0.75]	0.268	0.40
M10 $\log_{10}R_\infty$	0.016	0.129	0.99 [0.96, 1.00]	0.003	0.99
M10 α	0.031	0.069	0.89 [-0.13, 0.98]	0.008	0.55
M11 $\log_{10}Q$	0.075	0.336	0.95 [0.31, 0.99]	0.012	0.84
M11 α	0.018	0.069	0.95 [0.27, 0.99]	0.003	0.88
M11 $\log_{10}R_\infty$	0.025	0.132	0.99 [0.95, 1.00]	0.003	0.99

Table S20: Frequency-band design: percentage of synthetic spectra (20 per cell; 2.5% noise) in which $\log_{10}\tau$ was practically identifiable (nominal 95% profile interval closed and narrower than one decade), by design, f_c class and lower band limit ($f_{\max} = 100$ kHz; N frequencies at every $f_{\min}$).

Design	f_c (kHz)	10 kHz	3 kHz	1 kHz	300 Hz
Log grid, white, $N = 10$	10–30	95	100	100	100
	1–10	45	90	100	100
	0.1–1	0	15	60	100
Log grid, white, $N = 20$	10–30	95	100	100	100
	1–10	45	90	100	100
	0.1–1	0	20	55	90
Log grid, correlated, $N = 20$	10–30	90	100	100	100
	1–10	45	90	100	100
	0.1–1	5	25	65	95
Electrode pol. 10%, $N = 20$	10–30	95	100	100	100
	1–10	35	85	100	100
	0.1–1	5	25	55	90
Electrode pol. 50%, $N = 20$	10–30	70	80	40	0
	1–10	25	80	85	30
	0.1–1	5	15	35	80
Linear grid, white, $N = 20$	10–30	85	100	100	100
	1–10	45	85	100	100
	0.1–1	5	10	60	90
Model mismatch, $N = 20$	10–30	90	100	100	100
	1–10	20	60	85	100
	0.1–1	5	5	15	60

Table S21: Practical identifiability of $\log_{10}\tau$ (%) by $f_{\min}/f_c$, pooled over f_c classes, with the nominal threshold (3.84) / the median calibrated critical value of $\log_{10}\tau$ (6.38; Table S14).

Design	< 0.3	0.3–1	1–3	3–10	> 10
Log grid, white, $N = 10$	100 / 100	98 / 90	90 / 75	26 / 10	0 / 0
Log grid, white, $N = 20$	100 / 100	98 / 96	80 / 72	32 / 16	0 / 0
Log grid, correlated, $N = 20$	100 / 100	96 / 96	82 / 72	39 / 26	6 / 3
Electrode pol. 10%, $N = 20$	100 / 100	98 / 86	75 / 68	29 / 16	6 / 0
Electrode pol. 50%, $N = 20$	42 / 35	82 / 80	68 / 57	13 / 3	3 / 0
Linear grid, white, $N = 20$	100 / 100	94 / 90	80 / 75	26 / 6	3 / 0
Model mismatch, $N = 20$	100 / 100	84 / 80	40 / 28	0 / 0	6 / 6

Table S22: Mechanism of the VarPro gain: success (%) at 2,000 NFE in the same runs (20 spectra $\times$ 10 seeds) for full-parameter MS-TRF, VP-MS-TRF and VPinit-MS-TRF (full-parameter TRF started from VarPro-optimal linear coefficients), and median condition number of the full and reduced residual Jacobians at the best-known solutions.

Model	Runs	MS-TRF	VP-MS-TRF	VPinit-MS-TRF	κ full	κ reduced
M10	200	100	100	100	472	57
M12	200	64	98	67	2023	57

Table S23: Checks of the error model that do not rely on residuals alone (KK-consistent spectra unless stated). Upper rows: errors simulated under each covariance, scaled so that the expected residual sum of squares equals the observed one (100 replicates per spectrum): median implied SD of the fitted descriptor (number of the six recordings with repeated sweeps in which it exceeds the observed between-sweep SD) and coverage (%) of the calibrated intervals. [a]Median over the six recordings; includes drift and is therefore an upper bound for noise-induced variability. [b]Percentage of pairs of sweeps of the same recording whose estimates differ by less than $\sqrt{h_i^2 + h_j^2}$, h the interval half-width, with nominal / calibrated intervals; about 95% is expected for calibrated intervals without drift.

	$\log_{10} Q$	α	$\log_{10} R_\infty$	Cov. $\log_{10} Q$	Cov. α	Cov. $\log_{10} R_\infty$
Residual-based (used)	0.102 (4/6)	0.022 (4/6)	0.014 (3/6)	94	94	93
Projection-corrected	0.247 (6/6)	0.055 (6/6)	0.037 (4/6)	54	54	52
Observed between-sweep SD[a]	0.075	0.018	0.025			
Cross-sweep agreement[b], KK-consistent (31 pairs)	94 / 100	94 / 97	55 / 65			
Cross-sweep agreement[b], KK-inconsistent (34 pairs)	76 / 82	76 / 85	26 / 29			

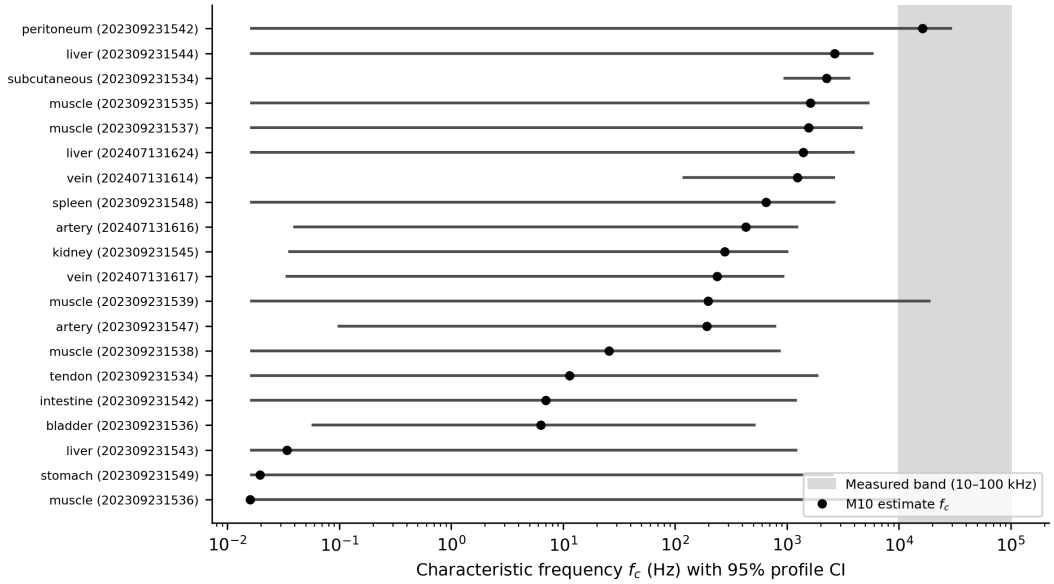


Figure S15: M10 estimates of the characteristic frequency f_c with nominal 95% profile-likelihood intervals relative to the measured band (shaded). Intervals extending to the left edge reach the search bound and are unbounded.

Table S24: Validation of Eqs. (8)–(9) on the band-design spectra (practical identifiability of $\log_{10}\tau$: closed profile interval narrower than one decade). Agreement: percentage of spectrum–band combinations classified identically by the closed-form width (Eq. (8)) or the exact linearized (Wald) width and by profile likelihood; false/missed: predicted identifiable but not observed / the reverse. Spearman and ratio: closed profiles only. Median ratio of predicted (Eq. (8) / Wald) to profile width by $f_{\min}/f_c$, white-noise designs: <1: 0.11 / 1.38 (n = 414); 1–3: 0.51 / 1.13 (n = 110); 3–10: 0.82 / 1.33 (n = 43); >10: 10.02 / 11.52 (n = 4). Median Γ (white log grid) for $f_{\min}$ of 10, 3, 1 and 0.3 kHz: 5.5, 21.2, 57.2, 161.7.

Design	n	Agreement Eq. (8) (%)	Agreement Wald (%)	False (%)	Missed (%)	Spearman	Median ratio
Log grid, white	480	92.5	93.8	5.6	1.9	0.77	0.23
Correlated noise	240	92.1	92.5	5.4	2.5	0.76	0.25
Electrode pol. 10%	240	89.2	91.2	7.9	2.9	0.53	0.18
Electrode pol. 50%	240	65.4	67.5	34.2	0.4	-0.03	0.08
Linear grid, white	240	91.2	91.7	7.1	1.7	0.86	0.16
Model mismatch	240	80.8	83.8	18.3	0.8	0.75	0.10

Table S25: Bias of the R–CPE–L descriptors fitted to Cole spectra: numerical refits (Table S16) and Eq. (10) (median over the 12 KK-consistent spectra). Median coefficients: $b_Q = 2.99$ decades and $b_\alpha = -0.58$ per unit $(f_{\min}/f_c)^{-\alpha}$.

$f_{\min}/f_c$	$\log_{10} Q$ refit	$\log_{10} Q$ Eq. (10)	α refit	α Eq. (10)
1000	0.028	0.028	-0.005	-0.005
333	0.059	0.059	-0.011	-0.011
100	0.133	0.134	-0.026	-0.026
33	0.279	0.284	-0.055	-0.056
10	0.617	0.643	-0.121	-0.125
3	1.216	1.347	-0.241	-0.262

Table S26: Closed-form band criterion applied to the 20 measured spectra. σ: residual SD per stacked relative component (Cole+L); Γ: Eq. (9) with $W^* = 1$ decade; detection limit $f_{\min}/\Gamma$ (Hz) with $z = 1.96$ and with the calibrated critical value; $\hat{f}_c$ and upper 95% bound of f_c (Hz) from the nominal profile; widths (decades) of the $\log_{10}\tau$ interval predicted by Eq. (8) at $\hat{f}_c$ and from the profile; identifiable predicted by the closed form / three-way rule (L likely, U uncertain-referred, N unlikely) / observed. f_{DL} (Hz): operational detection limit of each spectrum, nominal, computed with the pipeline of this repository on the public data (version 2 of the Zenodo record); for spectra 2, 3, 5, 6, 9–12 and 16 the public median spectra differ from those analysed here (Note S4.7), the other columns are unchanged. †KK-inconsistent.

Tissue	σ	α	Γ	f_c^*	f_c^* cal.	f_{DL}	$\hat{f}_c$	f_c upper	W Eq. (8)	W profile	Ident.
subcutaneous	0.021	0.66	9.31	1075	2165	2771	2267	3645	0.61	0.59	yes / U / yes
tendon	0.029	0.56	2.61	3832	6454	5371	11	1905	26.38	5.08	no / N / no
muscle†	0.069	0.73	1.79	5581	7865	7297	1625	5665	2.45	5.55	no / N / no
bladder	0.017	0.89	18.74	534	597	638	6	527	52.50	3.97	no / N / no
muscle†	0.164	0.63	0.29	35018	40521	>31623	0	10339	999.00	5.81	no / N / no
muscle†	0.055	0.72	2.46	4064	4860	6399	1568	5000	1.98	5.50	no / N / no
muscle	0.018	0.64	8.89	1125	1810	3811	26	885	11.20	4.75	no / N / no
muscle†	0.168	0.41	0.00	$>10^5$	$>10^5$	>31623	198	19162	46.15	6.08	no / N / no
intestine	0.018	0.68	6.12	1634	2160	4125	7	1227	41.08	4.89	no / N / no
peritoneum†	0.121	0.30	0.00	$>10^5$	$>10^5$	>31623	16275	30995	20.08	6.29	no / N / no
liver†	0.037	0.83	5.17	1934	2988	1687	0	1236	999.00	4.89	no / N / no
liver†	0.106	0.88	1.24	8053	10795	>31623	2674	5920	2.64	5.57	no / N / no
kidney	0.018	0.86	15.48	646	721	846	278	1030	2.06	4.47	no / N / no
artery	0.018	0.88	17.59	569	702	701	193	806	2.59	3.92	no / N / no
spleen	0.018	0.67	4.59	2179	3361	>31623	649	2711	2.26	5.23	no / N / no
stomach†	0.067	0.78	2.55	3924	6606	8177	0	2607	999.00	5.21	no / N / no
vein	0.018	0.69	8.57	1166	2306	3197	1240	2690	0.96	1.36	yes / U / no
artery	0.020	0.84	13.69	731	1016	1000	429	1307	1.57	4.53	no / N / no
vein	0.016	0.84	16.01	625	738	837	239	954	2.25	4.46	no / N / no
liver	0.022	0.67	3.57	2802	4306	>31623	1398	4038	1.59	5.40	no / N / no

Table S27: Separation of fitting uncertainty and between-sweep drift (six recordings with ≥ 2 sweeps in each group; REML). s: median fitting standard error (calibrated half-width/1.96); τ_d: drift SD with its 95% profile-likelihood interval (one-sided upper bound when $\hat{\tau}_d = 0$, reference 2.71) and the 95% posterior upper bound under half-normal priors (scale 0.05 / 0.2 decade for $\log_{10}$ parameters, 0.01 / 0.05 for α); half-widths: fitting $1.96s$ and reproducibility $1.96\sqrt{s^2+\tau_d^2}$.

Group	Parameter	Sweeps	s	$\hat{\tau}_d$	Profile 95% interval	Posterior upper 95%	Fitting half-width	Reprod. half-width
KK-consistent	$\log_{10} Q$	21	0.1143	0.0000	$\leq$0.050	0.053 / 0.067	0.224	0.224
	α	21	0.0248	0.0000	$\leq$0.011	0.011 / 0.015	0.049	0.049
	$\log_{10} R_\infty$	21	0.0112	0.0303	[0.020, 0.048]	0.046 / 0.048	0.022	0.063
KK-inconsistent	$\log_{10} Q$	22	0.1377	0.6693	[0.461, 1.040]	0.323 / 0.682	0.270	1.339
	α	22	0.0301	0.1577	[0.111, 0.241]	0.072 / 0.166	0.059	0.315
	$\log_{10} R_\infty$	22	0.0170	0.1098	[0.075, 0.183]	0.131 / 0.179	0.033	0.218

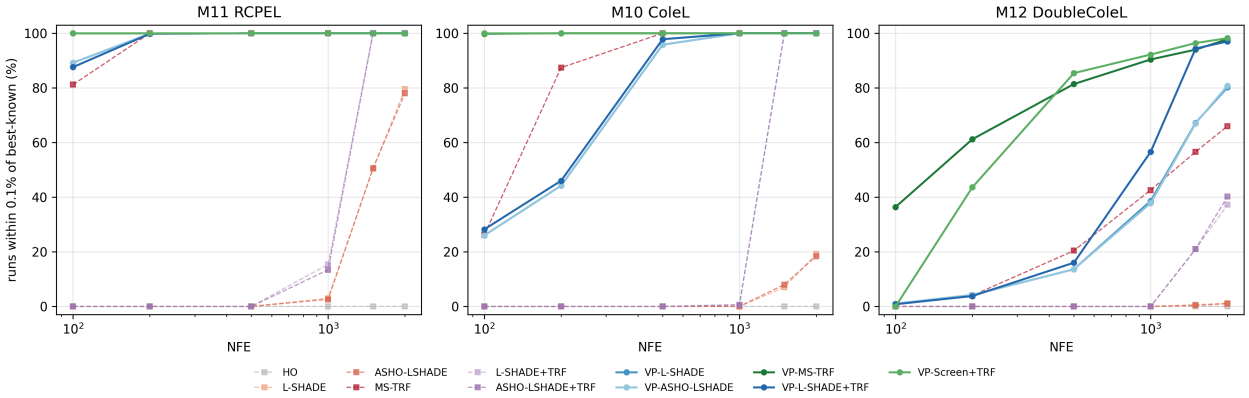


Figure S16: Percentage of runs within 0.1% of the best-known NRMSE versus NFE for the original (dashed) and variable-projection (solid) algorithms. VP-L-SHADE and VP-ASHO-LSHADE overlap.

Table S28: Time to solution of MS-TRF and VP-MS-TRF (20 spectra $\times$ 25 runs per model), measured in the same environment as the benchmark rerun of Table S30 (single-threaded Python 3.13, NumPy 2.5, SciPy 1.18, one core of an Intel Xeon at 2.1 GHz). Time to solution: wall-clock time until the NRMSE first came within 0.1% of the best-known value; unsuccessful runs were counted as infinite. n.r.: not reached within 2,000 NFE. [a]Among successful runs.

Model	Algorithm	Success (%)	Median NFE to solution[a]	Median time to solution, ms (IQR)	VP faster in paired runs (%)
M11	MS-TRF	100.0	76	7.3 (5.7–9.6)	99.8
	VP-MS-TRF	100.0	7	1.6 (1.2–2.1)	
M10	MS-TRF	100.0	133	12.0 (8.8–16.2)	93.0
	VP-MS-TRF	100.0	20	4.2 (2.7–6.5)	
M12	MS-TRF	65.2	768	128 (63.8–n.r.)	83.8
	VP-MS-TRF	96.8	132	27.8 (15.2–81.2)	

Table S29: Accuracy of the closed-form width of the $\log_{10}\tau$ interval by $r = (f_{\min}/f_c)^{-\alpha}$, the bound on the relative size of the neglected terms of Eq. (7): median ratio of predicted to profile-likelihood width (band-design spectra with closed profiles, white or correlated noise). First order: Eq. (8). Refined: exact derivative $-wu/(1+u)^2$ and delta method for $\log\tau = (\log Q - \log\eta)/\alpha$ including the covariances with Q and α (identical to the exact linearized width). Without coupling: refined derivative, η variance only. Coupling with Q and α widened the interval by a median factor of 1.22; classification agreement with profile likelihood: first order 92.1%, refined 92.9%.

r	n	First order	Refined	Without coupling
< 0.1	6	13.29	15.40	13.53
0.1–0.3	29	3.16	4.11	3.42
0.3–0.6	100	0.64	1.04	0.78
0.6–1	77	0.50	1.17	0.83
> 1 (f_c in band)	552	0.13	1.48	1.39

Table S30: Reproducibility of the benchmark across computing environments: success rate (%) at 2,000 NFE in the original benchmark (Table 9 of the main text; complete table in Table S53) and in a complete rerun with the same seeds in the timing environment of Table S28. Maximum absolute difference over all algorithms, models and checkpoints (500 and 2,000 NFE): 2.6 percentage points; no run found a value below the best-known reference. Differences arise from floating-point behaviour of different processors and library versions.

	M11		M10		M12	
Algorithm	Original	Rerun	Original	Rerun	Original	Rerun
HO	0.0	0.0	0.0	0.0	0.0	0.0
L-SHADE	79.6	79.8	19.2	19.0	1.2	1.2
ASHO-LSHADE	78.0	78.0	18.4	18.4	1.0	1.0
MS-TRF	100.0	100.0	100.0	100.0	66.0	65.2
L-SHADE+TRF	100.0	100.0	100.0	100.0	37.2	39.8
ASHO-LSHADE+TRF	100.0	100.0	100.0	100.0	40.2	39.8
VP-L-SHADE	100.0	100.0	100.0	100.0	80.2	80.2
VP-ASHO-LSHADE	100.0	100.0	100.0	100.0	80.8	80.8
VP-MS-TRF	100.0	100.0	100.0	100.0	97.8	96.8
VP-L-SHADE+TRF	100.0	100.0	100.0	100.0	97.0	97.2
VP-Screen+TRF	100.0	100.0	100.0	100.0	98.2	98.8

Table S31: Between-recording variability of the resistance and phase angle in recordings of the same medium with the same probe, animal and frequency (most stable window of each single-frequency recording; median over sweeps of the 100 kHz point of the frequency scans). Vessel recordings begin in the surrounding tissue and end in blood (separate needle advances); whether static repeat recordings involved re-insertion at a new site is not documented. SD: standard deviation of the recording medians; CV $= 10^{\mathrm{SD}} - 1$; short-term SD: median robust SD of $\log_{10} R$ within the stable window. Pooled: degrees-of-freedom-weighted SD with χ^2 95% interval.

Animal	Probe	Medium	kHz	n	Median R (Ω)	SD $\log_{10} R$	CV (%)	Short-term SD	SD phase (°)
1	monopolar	blood (artery)	100	4	1218	0.241	74	0.0022	5.5
1	monopolar	blood (vein)	100	4	1646	0.074	19	0.0014	1.5
1	white	blood (artery)	100	3	1616	0.148	41	0.0068	3.6
1	white	blood (vein)	100	2	2211	0.017	4	0.0052	1.0
1	yellow	blood (vein)	100	6	3053	0.187	54	0.0062	3.1
1	yellow	liver	100	2	2185	0.112	29	–	1.1
1	yellow	muscle	100	7	2680	0.078	20	0.0011	4.4
1	yellow	skin	100	2	2627	0.048	12	0.0006	8.4
1	yellow	subcutaneous	100	3	3623	0.199	58	0.0013	13.6
1	yellow	tendon	100	2	4519	0.046	11	0.0019	14.8
2	yellow	blood (artery)	20	2	2796	0.004	1	0.0012	0.0
2	yellow	blood (vein)	20	4	3929	0.060	15	0.0049	8.2
2	yellow	vein (scans)	100	2	2830	0.344	121	–	4.2
Pooled									
blood, bipolar (same probe)				17 (df 12)		0.138 [0.099, 0.228]	37		4.8
blood, monopolar				8 (df 6)		0.179 [0.115, 0.393]	51		4.0
tissue, yellow bipolar 100 kHz				18 (df 12)		0.145 [0.104, 0.239]	40		8.1
within recording, between sweeps, KK-consistent (R at 100 kHz)				6 (df 15)		0.034 [0.025, 0.053]	8		1.2
within recording, between sweeps, all (R at 100 kHz)				12 (df 31)		0.108 [0.087, 0.143]	28		–
within recording, short-term (stable window)				36 (df –)		0.002	0.4		–

Table S32: Probe effect: mean $\log_{10} R$ at 100 kHz with the second bipolar probe (white) minus that with the scan probe (yellow), first animal. Differences in blood, tendon and subcutaneous tissue exceed the between-recording SD (Table S31), so absolute impedance descriptors are probe-specific.

Medium	n yellow	n white	$\Delta \log_{10} R$	Difference (%)
tendon	2	1	-0.28	-48
subcutaneous	3	1	-0.38	-59
bladder	1	1	+0.06	+16
intestine	1	1	-0.11	-22
blood	6	5	-0.20	-37

Table S33: Descriptor-level between-recording SD from recordings of the same tissue in the same animal (all: liver $\times 2$ and muscle $\times 5$, first animal; vein $\times 2$, second animal; KK-consistent: the vein pair only). Within-recording SD: root mean of fitting variance (calibrated half-width/1.96) plus drift variance (Table S27). Excess: $\sqrt{\max(\mathrm{SD}^2 - \mathrm{within}^2, 0)}$; upper bound from the χ^2 95% upper limit of the SD. In the pooled set the within-recording variance of the KK-inconsistent recordings exceeds the between-recording variance, so no excess is estimable; with one degree of freedom the KK-consistent bound is uninformative.

Parameter	Set	df	Between-recording SD	Within-recording SD	Excess	Excess, upper 95% bound
$\log_{10} Q$	all	6	0.423	0.692	0.000	0.62
$\log_{10} Q$	KK-consistent	1	0.289	0.102	0.271	9.23
α	all	6	0.113	0.158	0.000	0.19
α	KK-consistent	1	0.109	0.022	0.106	3.46
$\log_{10} R_\infty$	all	6	0.360	0.787	0.000	0.10
$\log_{10} R_\infty$	KK-consistent	1	0.317	0.032	0.315	10.11

Table S34: Scale test. A pure scale change of the impedance ($Z \to kZ$, e.g. a change of cell constant) shifts $\log_{10} R_\infty$ and $\log_{10} Q$ by equal and opposite amounts (slope -1), leaves α and the phase unchanged and cancels in $\log_{10}(R_\infty Q)$. Slope: regression of within-group deviations of $\log_{10} Q$ on those of $\log_{10} R_\infty$. Only the pooled set, dominated by KK-inconsistent muscle recordings, behaves partly like a scale change; the KK-consistent replicate pair and the between-sweep changes do not, consistent with the phase changes in Table S31. Transfer slope from $\log_{10} R(100\ \mathrm{kHz})$ to $\log_{10} R_\infty$ within KK-consistent recordings: 0.96 ($r = 0.98$, 21 sweeps).

Set	SD $\log_{10} R_\infty$	SD $\log_{10} Q$	SD $\log_{10}(R_\infty Q)$	SD α	Slope
all replicate groups	0.360	0.423	0.301	0.113	-0.84
KK-consistent replicate groups	0.317	0.289	0.606	0.109	+0.91
within recording, between sweeps (KK-consistent)	0.029	0.092	0.111	0.022	+1.81

Table S35: Repeatability and reproducibility limits (ISO 5725 form, $r = 2.77\, s_w$, $R = 2.77\sqrt{s_w^2 + s_b^2}$) and tissue discrimination. s_w: within-recording SD (fitting and drift, KK-consistent recordings); s_b: between-recording component (upper 95% bound in parentheses), for R_∞ from the bipolar blood recordings transferred with the slope of Table S34, for Q and α from the single KK-consistent replicate pair (point estimates only indicative; bounds uninformative). Discrimination: absolute differences between tissue means of the 10 tissues with KK-consistent spectra (45 pairs) and the percentage of pairs exceeding r and R.

Parameter	s_w	r	s_b	R	Source of s_b	Median $\lvert\Delta\rvert$	$> r$ (%)	$> R$ (%)
$\log_{10} R_\infty$	0.032	0.090	0.128 (0.22)	0.37 (0.60)	blood recordings	0.317	84	42 (11)
$\log_{10} Q$	0.114	0.317	0.271 (9.23)	0.81 (25.57)	one replicate pair	0.446	69	18 (0)
α	0.025	0.069	0.106 (3.46)	0.30 (9.60)	one replicate pair	0.112	71	2 (0)

Table S36: Main results recomputed per animal (animal, recording subset and date are confounded: first animal = subset A, second = subset D). The detection limit is the per-spectrum median and differs from the pooled value in the main text.

Quantity	First animal	Second animal
Spectra (KK-consistent)	16 (8)	4 (4)
M12 success, MS-TRF / VP-MS-TRF (%)	74 / 98	34 / 98
Spectra with VP gain / spectra that differ	11/11	4/4
$\log_{10}\tau$ practically identifiable	1/16	0/4
Band criterion agrees with profile likelihood	16/16	3/4
Median Γ	3.6	11.1
Median detection limit, nominal / calibrated (kHz)	3.0 / 4.1	0.9 / 1.7
Closed calibrated intervals Q, α, R_∞ (KK, %)	100, 100, 100	100, 100, 100
Median calibrated half-width $\log_{10} Q$ / α (KK)	0.21 / 0.045	0.21 / 0.044
Artery, one KK-consistent scan per animal: second minus first animal		
α: difference (fitting SE)	-0.036 (0.023); r = 0.069	
$\log_{10} Q$: difference (fitting SE)	+0.317 (0.108); r = 0.317	
$\log_{10} R_\infty$: difference (fitting SE)	-0.048 (0.010); r = 0.090	

Table S37: Cluster-aware inference for the benchmark. Upper block: the four confirmatory hypotheses with Holm-adjusted p-values (H1, H4: exact one-sided sign-flip permutation of the means of the 14 tissue-within-animal clusters; H2, H3: two one-sided tests at ± 5 percentage points with the two-stage cluster bootstrap, 90% interval shown; H4 averages $\log_{10} Q$, α and $\log_{10} R_\infty$ under the empirical covariance). Lower block: 95% intervals of all benchmark contrasts at 2,000 NFE when resampling spectra, clusters, or clusters and then spectra within clusters, and from a run-level linear probability model with cluster-robust (CR1) standard errors. Clustering by tissue only (12 clusters) gave similar intervals (results files).

H	Hypothesis	Estimate	Interval	p	p_{Holm}	Supported
H1	VP-MS-TRF > MS-TRF, M12 success	+31.8 pp	19.7 to 47.7	0.0002	0.0007	yes
H2	ASHO-LSHADE equivalent to L-SHADE (± 5 pp), M12	-0.2 pp	-1.1 to 0.5 (90%)	$< 10^{-4}$	$< 10^{-4}$	yes
H3	ASHO-LSHADE+TRF equivalent to L-SHADE+TRF (± 5 pp), M12	+3.0 pp	-0.3 to 5.9 (90%)	0.124	0.124	no
H4	calibrated > nominal coverage (empirical Σ)	+10.8 pp	9.2 to 12.5	0.0005	0.0010	yes

All contrasts: per-spectrum success difference (pp), 95% intervals by resampling unit

Model	Contrast	Estimate	Spectra	Clusters (14)	Two-stage	Cluster-robust LPM
M12	VP-MS-TRF - MS-TRF	+31.8	20.4 to 43.6	20.8 to 47.5	19.7 to 47.7	18.5 to 45.1
M12	VP-L-SHADE+TRF - L-SHADE+TRF	+59.8	48.2 to 70.8	46.9 to 72.0	45.6 to 71.8	47.6 to 72.0
M10	VP-L-SHADE - L-SHADE	+80.8	75.8 to 85.6	76.2 to 85.6	75.6 to 86.1	76.3 to 85.3
M12	VP-L-SHADE - L-SHADE	+79.0	66.8 to 89.6	61.1 to 90.1	61.0 to 90.1	64.0 to 94.0
M12	ASHO-LSHADE - L-SHADE	-0.2	-1.2 to 0.6	-1.3 to 0.5	-1.4 to 0.6	-1.1 to 0.7
M12	ASHO-LSHADE+TRF - L-SHADE+TRF	+3.0	0.2 to 6.4	-0.8 to 5.6	-0.8 to 6.4	-0.4 to 6.4
M11	ASHO-LSHADE - L-SHADE	-1.6	-4.2 to 0.8	-4.6 to 0.6	-4.9 to 1.0	-4.3 to 1.1
M10	ASHO-LSHADE - L-SHADE	-0.8	-3.6 to 1.8	-3.2 to 2.1	-3.6 to 2.3	-3.4 to 1.8

Table S38: Robustness of the between-recording SD of $\log_{10} R$ (Table S31): classical pooled SD, range when one group is left out, and a pooled median-absolute-deviation (MAD) estimate; ratios are relative to the between-sweep SD within KK-consistent recordings (0.034 decade). Between-recording variability exceeds within-recording variability in every analysis, but its size depends on a few discordant recordings; on the R_∞ scale the reproducibility limit is 0.37 decade with the classical and 0.15 decade with the MAD estimate.

Set	Groups	df	SD	Ratio	SD, leave one group out	Ratio	MAD SD	Ratio
blood, bipolar	5	12	0.138	4.0	0.088–0.156	2.6–4.6	0.057	1.7
tissue, scan probe	6	12	0.145	4.2	0.110–0.189	3.2–5.5	0.075	2.2
all same-probe groups	13	30	0.150	4.4	0.136–0.163	4.0–4.8	0.079	2.3

Table S39: Planning a replicate study: reproducibility limit of tissue means of $\log_{10} R_\infty$ when n placements per tissue are averaged, $2.77\sqrt{(s_w^2 + s_b^2)/n}$ with the classical estimates of Table S35, and the percentage of the 45 observed tissue pairs (KK-consistent spectra) whose difference exceeds it.

Placements per tissue n	Limit (decade)	Pairs exceeding (%)
1	0.365	42
2	0.258	60
3	0.211	69
4	0.183	71
6	0.149	76
8	0.129	78

Table S40: Coverage (%) of nominal and calibrated 95% profile intervals (12 KK-consistent spectra $\times$ 100 replicates) and measured-spectrum statistics, with two-stage cluster-bootstrap 95% intervals (clusters: tissue within animal).

Condition	Parameter	Nominal	Calibrated
empirical Σ	$\log_{10} Q$	83.8 (81.4 to 85.9)	93.2 (91.4 to 94.8)
empirical Σ	α	82.9 (80.6 to 84.9)	92.9 (90.7 to 94.8)
empirical Σ	$\log_{10} R_\infty$	81.1 (78.6 to 83.4)	94.0 (92.3 to 95.6)
$\rho_f + 0.2$	$\log_{10} Q$	77.0 (75.3 to 78.8)	88.3 (86.0 to 90.5)
$\rho_f + 0.2$	α	75.7 (74.0 to 77.5)	88.3 (85.9 to 90.8)
$\rho_f + 0.2$	$\log_{10} R_\infty$	71.0 (69.9 to 71.9)	88.2 (86.2 to 90.0)
AR(1)	$\log_{10} Q$	69.5 (66.0 to 72.6)	82.8 (79.7 to 85.5)
AR(1)	α	69.7 (66.5 to 72.5)	83.8 (81.2 to 86.3)
AR(1)	$\log_{10} R_\infty$	67.3 (65.7 to 69.2)	84.8 (82.9 to 86.6)
Spearman, detection limit vs upper f_c bound		0.89 (0.60 to 0.98)	
Median Γ, measured spectra		4.9 (2.5 to 13.7)	

Table S41: Precision of an estimated standard deviation: 95% χ^2 interval as a multiple of the estimate, by degrees of freedom. An upper limit within 1.5 times the estimate requires about 20 degrees of freedom, for example ten tissues with three placements each; the present KK-consistent replicate data for Q and α have one.

df	Lower factor	Upper factor
1	0.45	31.91
2	0.52	6.28
5	0.62	2.45
10	0.70	1.75
20	0.77	1.44
30	0.80	1.34
50	0.84	1.24
100	0.88	1.16

Table S42: Complete definitions of all equivalent-circuit models. All models were fitted to the weighted relative residual of Eq. (4) with weights $1/\max(|Z_k^{\mathrm{obs}}|, 1\,\Omega)$. Positive parameters were optimized as $\log_{10}$ of their value in the units shown, within the bounds given (search bounds); exponents were optimized on a linear scale. In the variable-projection algorithms, the linear coefficients (R terms, $1/Q$, L) were bounded on the linear scale by the same limits (10^{lower} to 10^{upper}). $Z_{\mathrm{CPE}}(Q,\alpha) = 1/[Q(j\omega)^{\alpha}]$; $A \parallel B = (1/A + 1/B)^{-1}$; $Z_C = 1/(j\omega C)$. Units: R in Ω, C in F, Q in S s$^{\alpha}$, τ in s, L in H. M1–M11 were screened (Table S44); M10, M11 and M12 were benchmarked; M13 tested the limit of variable projection. [a]M3, M4 and M6 are mathematically equivalent model classes (an $R \parallel$CPE element is a Cole element with $\Delta R = R_1$, $\tau^{\alpha} = R_1 Q$) and gave identical fits; they were retained to document the original screening. [b]R_0 and R_1 enter only through their sum, so one parameter is redundant; the redundancy does not affect the fit or the AICc comparison by more than the penalty of one parameter.

Model	$Z(\omega)$	k	Parameters: $\log_{10}$ search bounds (exponents: linear)
M1 Series RC	$R_s + Z_C$	2	R_s [−3, 7]; C [−12, −1]
M2 Fricke–Morse (Debye form)	$R_s + (R_1 \parallel Z_C)$	3	R_s [−3, 7]; R_1 [−3, 8]; C [−12, −1]
M3 Single Cole[a]	$R_\infty + \Delta R/[1 + (j\omega\tau)^{\alpha}]$	4	R_∞ [−3, 7]; ΔR [−3, 8]; τ [−7, 1]; α [0.3, 1]
M4 Randles–CPE[a]	$R_s + (R_1 \parallel Z_{\mathrm{CPE}}(Q,\alpha))$	4	R_s [−3, 7]; R_1 [−3, 8]; Q [−12, 0]; α [0.3, 1]
M5 Double RC	$R_s + (R_1 \parallel Z_{C_1}) + (R_2 \parallel Z_{C_2})$	5	R_s [−3, 7]; $R_{1,2}$ [−3, 8]; $C_{1,2}$ [−12, −1]
M6 Fricke–CPE[a]	$R_s + (R_1 \parallel Z_{\mathrm{CPE}}(Q,\alpha))$	4	as M4
M7 Double CPE	$R_s + \sum_{i=1}^{2} (R_i \parallel Z_{\mathrm{CPE}}(Q_i,\alpha_i))$	7	R_s [−3, 7]; $R_{1,2}$ [−3, 8]; $Q_{1,2}$ [−12, 0]; $\alpha_{1,2}$ [0.3, 1]
M8 Double Cole	$R_\infty + \sum_{i=1}^{2} \Delta R_i/[1 + (j\omega\tau_i)^{\alpha_i}]$	7	R_∞ [−3, 7]; $\Delta R_{1,2}$ [−3, 8]; $\tau_{1,2}$ [−7, 1]; $\alpha_{1,2}$ [0.3, 1]
M9 Cole + electrode CPE[b]	$R_0 + R_1 + \Delta R/[1 + (j\omega\tau)^{\alpha}] + Z_{\mathrm{CPE}}(Q_e,\alpha_e)$	7	R_0, R_1 [−3, 7]; ΔR [−3, 8]; τ [−7, 1]; Q_e [−12, 0]; α, α_e [0.3, 1]
M10 Cole+L	Eq. (1)	5	R_∞ [−3, 7]; ΔR [−3, 8]; τ [−7, 1]; α [0.3, 1]; L [−9, 0]
M11 R–CPE–L	Eq. (3)	4	R_∞ [−3, 7]; Q [−12, 0]; α [0.3, 1]; L [−9, 0]
M12 Double Cole+L	M8 $+ j\omega L$	8	as M8; L [−9, 0]
M13 Nested	$\left[Z_{10}^{-1} + Q_p(j\omega)^{\beta}\right]^{-1} + j\omega L$, Z_{10} without L	7	R_∞ [−3, 7]; ΔR [−3, 8]; τ [−7, 1]; α [0.3, 1]; Q_p [−14, −6]; β [0.5, 1]; L [−9, 0]

Table S43: High-frequency correction variants of M11 used in the sensitivity analysis (Table S17), with their bounds; $Z_{\mathrm{RC}} = R_\infty + 1/[Q(j\omega)^{\alpha}]$ with R_∞, Q, α bounded as in M11. In T2 the factor $e^{-j\omega t_d}$ is a phase shift linear in frequency; $t_d > 0$ is a conventional delay (phase lag) and $t_d < 0$ a phase lead. The fitted t_d was negative (median -0.55 μs, all spectra; -0.50 μs, KK-consistent spectra), so T2 is best read as an empirical linear phase correction rather than as a physical delay.

Treatment	$Z(\omega)$	Band	Additional parameter and bounds
T0 (reference)	$Z_{\mathrm{RC}} + j\omega L$	10–100 kHz	L: 10^{-9}–1 H
T1	Z_{RC}	10–80 kHz	–
T1b	$Z_{\mathrm{RC}} + j\omega L$	10–80 kHz	L: 10^{-9}–1 H
T2 linear phase correction	$Z_{\mathrm{RC}}\, e^{-j\omega t_d}$	10–100 kHz	t_d: −5 to 5 μs
T3 constant phase offset	$Z_{\mathrm{RC}}\, e^{j\varphi_0}$	10–100 kHz	φ_0: −0.8 to 0.8 rad
T4 generalized term	$Z_{\mathrm{RC}} + K(j\omega)^{\nu}$	10–100 kHz	K: 10^{-12}–10^{6}; ν: 0.3–1

Table S44: Circuit representation with the 12 KK-consistent spectra as the primary set and all 20 spectra as a robustness analysis. (a) Weighted-AICc screening: median NRMSE, AICc/BIC wins and mean AICc rank. (b) M10 versus M11. (c) Descriptors. In the KK-consistent spectra, M10 predicted the band-edge frequencies better than M11 in 10 of 12 spectra (Wilcoxon $p = 0.05$), a hint that the Cole model extrapolates slightly better although it does not fit the band better.

Model	k	KK-consistent (12): NRMSE, AICc/BIC wins, rank			All (20)		
(a) Screening, five best models							
M11 R–CPE–L	4	0.0227	10 / 10	1.17	0.0321	18 / 17	1.10
M10 Cole+L	5	0.0219	2 / 2	1.83	0.0262	2 / 3	2.15
M2 Fricke–Morse	3	0.0916	0 / 0	3.42	0.1204	0 / 0	3.30
M6 Fricke–CPE	4	0.0916	0 / 0	5.00	0.1204	0 / 0	5.15
M4 Randles–CPE	4	0.0916	0 / 0	5.42	0.1204	0 / 0	5.35
(b) M10 versus M11 (number of spectra; KK-consistent / all)							
M11 preferred, AICc, independent, 10–100 kHz				10			18
M11 preferred, AICc, independent, 10–80 / 20–100 kHz				11 / 12			19 / 19
M11 preferred, AICc, GLS, 10–100 kHz				12			20
Bootstrap LR rejects M11 ($p < 0.05$), independent / GLS				2 / 0			2 / 0
Median ΔAICc (M10 − M11), independent / GLS				2.62 / 3.28			2.65 / 3.23
LOFO interior: M11 better (Wilcoxon p)				9 (0.34)			14 (0.11)
LOFO band edges: M11 better (Wilcoxon p)				2 (0.05)			9 (0.76)
(c) M11 descriptors, median (KK-consistent / all)							
R_∞ (Ω); Q (S s$^\alpha$); α; L (mH)				2211 / 1888; 7.4 / 7.1 $\times 10^{-8}$; 0.69 / 0.70; 1.14 / 1.17			

Table S45: High-frequency-correction sensitivity in the 12 KK-consistent spectra: median NRMSE and median absolute change of the M11 descriptors relative to T0; extra parameter: L (H), t_d (μs), φ_0 (rad) or ν. Last two columns: all 20 spectra.

Treatment	NRMSE	$\lvert\Delta\log_{10} R_\infty\rvert$	$\lvert\Delta\log_{10} Q\rvert$	$\lvert\Delta\alpha\rvert$	Extra	All 20: $\lvert\Delta\log_{10} Q\rvert$, $\lvert\Delta\alpha\rvert$	
T0 series L	0.023	0.000	0.000	0.000	0.00114	0.000	0.000
T1 none, 10–80 kHz	0.095	0.059	0.525	0.122	–	0.530	0.124
T1b L, 10–80 kHz	0.023	0.008	0.050	0.011	0.00134	0.051	0.011
T2 linear phase	0.025	0.031	0.053	0.012	-0.502	0.053	0.010
T3 phase offset	0.053	0.018	0.692	0.135	0.182	0.728	0.143
T4 $K(j\omega)^\nu$	0.022	0.028	0.082	0.019	0.916	0.082	0.019

Table S46: Residual-based error model estimated from all 51 sweeps (published; used for calibration in the main text) and from the 27 sweeps of the KK-consistent recordings only. The KK-only model is close to independent errors with a larger imaginary component; the correlated structure of the published model comes mainly from KK-inconsistent recordings. σ_0 is immaterial for calibration because the covariance is rescaled to the observed RSS of each spectrum.

Parameter	All 51 sweeps	KK-consistent only
σ_0	0.110	0.012
γ	0.009	-0.063
ρ_f	0.180	-0.057
ρ_{RI}	-0.194	0.300
κ	0.459	1.904
Sweeps (recordings)	51 (20)	27 (12)

Table S47: Coverage (%) of nominal intervals, of intervals calibrated with the KK-only error model and of the published calibrated intervals (critical values from the model estimated from all 51 sweeps); 12 KK-consistent spectra × 100 replicates. (a) Errors drawn from the KK-only model: nominal intervals are nearly adequate and the published calibration is conservative. (b) The reverse case: errors from the published model. The choice is asymmetric: intervals calibrated with the published model stay at or above the nominal level under either error structure (93–99%), whereas intervals calibrated with the KK-only model under-cover when the errors are correlated (78–89%). The size of the nominal under-coverage reported in the main text is conditional on the error model.

Parameter	Nominal	Calibrated, KK-only Σ	Calibrated, published Σ	Median c^* KK-only	Median c^* published
(a) Errors drawn from the KK-only model					
α	92.2	94.0	97.7	4.54	7.23
$\log_{10} Q$	90.6	93.6	97.1	4.93	6.71
$\log_{10} R_\infty$	94.8	93.4	99.3	3.53	7.75
(b) Errors drawn from the published model (all 51 sweeps)					
α	82.9	85.9	92.9		
$\log_{10} Q$	83.8	88.9	93.2		
$\log_{10} R_\infty$	81.1	77.5	94.0		

Table S48: Closed-form criterion by $r = (f_{\min}/f_c)^{-\alpha}$ for all band-design spectra (white noise, correlated noise and linear grid), complementing Table S29, which is restricted to closed profiles. For $r < 0.1$ only 7% of profiles closed; in that regime the predicted first-order and exact linearized (Wald) widths are many decades and correctly classify $\log_{10} \tau$ as not practically identifiable (agreement 97%), whereas the few profiles that happened to close are much narrower than any local (Fisher) approximation, because the likelihood is far from quadratic and the closed profiles are a selected subset. The width ratios of Table S29 for $r < 0.1$ therefore reflect non-local profiles, not a failure of the expansion (whose refined form equals the Wald width). Classification errors concentrate in the transition range $r = 0.3$–0.6.

r	Spectra	Closed (%)	Median first-order width	Median Wald width	Predicted not ident. (%)	Profile not ident. (%)	Agreement (%)
<0.1	92	7	16.91	20.64	100	97	97
0.1-0.3	116	25	2.17	2.88	92	83	82
0.3-0.6	120	83	0.40	0.71	3	38	66
0.6-1	80	96	0.15	0.35	0	5	95
>1	552	100	0.01	0.10	0	1	99

Table S49: Hierarchy of variability of the resistance and phase angle (100 kHz; 20 kHz for the blood recordings of the second animal). Between-recording rows: pooled over groups of recordings of the same medium with the same probe and animal (Table S31). CV $= 10^{\mathrm{SD}} - 1$. Interval: χ^2 95% interval of the pooled SD.

Source of variability	Recordings (df)	SD $\log_{10} R$ [95% interval]	CV (%)	SD phase (°)
Short term, within stable window	36	0.002	0.4	–
Between sweeps, KK-consistent recordings	6 (15)	0.034 [0.025, 0.053]	8	1.2
Between sweeps, all recordings	12 (31)	0.108 [0.087, 0.143]	28	–
Between recordings, blood, bipolar probes	17 (12)	0.138 [0.099, 0.228]	37	4.8
Between recordings, blood, monopolar probe	8 (6)	0.179 [0.115, 0.393]	51	4.0
Between recordings, tissue, scan probe	18 (12)	0.145 [0.104, 0.239]	40	8.1
Between probes, same medium (5 media)	–	$\lvert\Delta\rvert$ 0.06–0.38	15–142	–

Supplementary Note S1: Additional quantitative results

This note collects quantitative results that support the main text but were moved here for readability. Section numbers refer to the main text.

S1.1 Measurement quality and error structure (Sect. 3.1)

The imaginary-to-real standard-deviation ratio of the error model was $\kappa = 0.46$, with no clear frequency trend in the standard deviations. Refitting absorbed a median 85% of the deviation energy between successive sweeps (interquartile range 41–93%), so most variation between sweeps appeared as changes in fitted parameters rather than as misfit.

S1.2 Circuit representation (Sect. 3.2)

In all 20 spectra, M11 ranked first by AICc in 18 and by BIC in 17. The bootstrap likelihood-ratio test rejected M11 in 2 of 12 KK-consistent spectra under the independent likelihood (the same 2 of 20 overall) and in none under GLS. In leave-one-frequency-out prediction, M11 was better at interior frequencies in 9 of 12 spectra ($p = 0.34$), whereas M10 predicted the band edges better in 10 of 12 ($p = 0.05$; Table S44; the corresponding counts for all 20 spectra are in Table S18). The four measurement-chain corrections that fitted equally well (NRMSE 0.022–0.025 in the KK-consistent spectra) changed $\log_{10} Q$ by a median of 0.05–0.08 decade and α by 0.011–0.019 relative to series L, and $\log_{10} R_\infty$ by up to 0.03 decade (0.05 in all spectra); omitting the correction or using a constant phase offset shifted $\log_{10} Q$ by 0.52 and 0.69 decade (Tables S17 and S45). The linear phase correction converged to negative t_d (median $-0.50\ \mu$s), a phase lead rather than a physical delay (Table S43).

S1.3 Calibrated uncertainty and identifiability (Sect. 3.3)

Calibrated GLS intervals covered 93–94% and the double bootstrap 92–93%. Median calibrated critical values were 6.8 for $\log_{10} Q$ and 7.4 for α. Under strongly correlated noise, nominal coverage was 67–70%. At a stricter threshold of 0.5 decade (0.1 for α), calibrated Q and α were identifiable in only 8 and 6 of 12 KK-consistent spectra; at 1.5 decades and with wider bounds, all three R–CPE–L descriptors remained identifiable in all 12 (Table S15). Profiles are shown in Fig. S14.

S1.4 Closed-form band criterion and band design (Sects. 3.4 and 3.5)

Within the intended range ($r = 0.3$–1), the first-order width was within a factor of two of the profile width (median ratio 0.50–0.64) and the refined width within 4–17%; the coupling with Q and α widened the interval by a median of 22% (Table S29). For $r < 0.1$ only 7% of profiles closed, and those that did were 13–15 times narrower than the predicted widths (Table S48). With the calibrated critical value (6.38), the band rule was unchanged (86–100%, 68–75% and at most 26% identifiable for $f_{\min}/f_c < 1$, 1–3 and 3–10; Table S21).

S1.5 Reliability of the inversion (Sect. 3.8)

With 10,000 NFE, L-SHADE reached 97.5–100% success on M10 and M11. The best VarPro solution matched the best-known value for every spectrum and circuit (relative difference below 10^{-10}), and the M12 gain of VP-MS-TRF over MS-TRF was positive in all 15 spectra in which the methods differed. On M12, the memetic variants succeeded in 37.2–40.2% and the L-SHADE family in at most 1.4% of runs. The M12 gain of VP-MS-TRF over MS-TRF held in both recording subsets (Table S4), did not depend on the success tolerance (Table S11) and stayed between 29.1 and 33.5 percentage points when any spectrum was removed. VarPro was faster to solution despite its higher cost per evaluation (Table S28): median 1.6 versus 7.3 ms on M11 and 4.2 versus 12.0 ms on M10 (28 versus 128 ms on M12), with VP-MS-TRF faster in 83.8–99.8% of paired runs.

S1.6 Reproducibility across sweeps, recordings and animals (Sect. 3.9)

Between sweeps, the median standard deviation of $\log_{10} \Delta R$ was 0.63 decade; ICCs were 0.48 ($\log_{10} \tau$) and 0.43 ($\log_{10} \Delta R$) versus 0.95, 0.95 and 0.99 for $\log_{10} Q$, α and $\log_{10} R_\infty$ (Table S19); reading-level resampling understated the between-sweep variability two- to eightfold (Table S19; ranking stability in Table S6). Drift in $\log_{10} Q$ and α was bounded above by 0.050 decade and 0.011 (Table S27). In KK-inconsistent recordings, all descriptors drifted, by five to seven times their fitting standard errors (Table S27).

Within a stable window, single-frequency noise was 0.4%. Between recordings of the same medium with the same probe, the phase varied by 4–8° against 1.2° between sweeps, so the effect was not a pure scale change such as a change of cell constant. The between-to-within ratio was 2.6–5.5 when one group was left out and 1.7–2.2 with a

pooled median-absolute-deviation estimator (Table S38). In the one KK-consistent replicate pair (two vein scans), $\log_{10} Q$, α and $\log_{10} R_\infty$ differed by 0.41 decade, 0.15 and 0.45 decade, and $\log_{10}(R_\infty Q)$, which cancels a scale change, did not differ less. For R_∞, the robust between-recording component was 0.04 decade (upper 95% bound 0.22); 84% of the 45 pairs of tissues with KK-consistent spectra differed by more than the repeatability limit but only 42–76% (11% at the bound) by more than the reproducibility limit. With four placements per tissue, averaging would lower the classical limit to 0.18 decade (Table S39). In each animal (Table S36), the VarPro gain on M12 held in 11 of 11 and 4 of 4 spectra in which the methods differed, τ was non-identifiable in 15 of 16 and 4 of 4 spectra, and the band criterion agreed with profile likelihood in 16 of 16 and 3 of 4.

Supplementary Note S2: Additional methods

S2.1 Measurement-chain corrections (Sect. 2.2)

Besides the series inductance of M11, five alternative high-frequency corrections were fitted (Table S43; T0 denotes the reference series L on 10–100 kHz): (T1) no correction, fitted on 10–80 kHz; (T1b) series L, fitted on 10–80 kHz; (T2) a linear phase factor $e^{-j\omega t_d}$, the form of the usual time-delay compensation, with t_d of either sign; (T3) a constant phase offset $e^{j\varphi_0}$; and (T4) a term $K(j\omega)^\nu$ with $\nu \in [0.3, 1]$, which contains the inductance as $\nu = 1$.

S2.2 Error model (Sect. 2.3)

The 95% intervals of the error-model parameters ψ were obtained by resampling recordings (200 replicates). Ledoit–Wolf shrinkage of the sample covariance was used as a check. A projection-corrected likelihood, which accounts for the fact that residuals are projections of the errors, was also fitted but was weakly identified (Table S13).

S2.3 Between-recording variability (Sect. 2.9)

The 35 single-frequency recordings were made at 100 kHz (20 kHz in the second animal) with the scan probe, a second bipolar probe and a monopolar probe, and were not used for fitting. Vessel recordings begin in the surrounding tissue and end in blood, so each is a separate needle advance; for static repeats, re-insertion is not documented. Each recording was summarized by the median resistance and phase in its most stable window (20% of readings), and the 100 kHz point of the scans was added. Pooled standard deviations with χ^2 intervals were computed within groups of the same medium, probe, animal and frequency (25 blood and 18 tissue recordings). Robustness was assessed by leaving out one group at a time and by a pooled median-absolute-deviation estimator. The main results were also recomputed for each animal.

S2.4 Benchmark and statistical details (Sects. 2.4–2.6)

The best-known value of each spectrum and circuit was the best of all runs, refined by 30 multistart fits. Timing used single-threaded Python 3.13, NumPy 2.5 and SciPy 1.18 on one Intel Xeon core at 2.1 GHz. Equivalence hypotheses (H2, H3) were tested by two one-sided tests. M10 and M11 were compared by AICc under the independent and GLS likelihoods, by leave-one-frequency-out prediction (interior frequencies and band edges) and by a parametric-bootstrap likelihood-ratio test with 100 replicates per spectrum, because M11 lies on the boundary of M10. As a second check of the error model against the repeated sweeps, the descriptor variability implied by a candidate covariance was compared with the observed between-sweep variability.

Supplementary Note S3: Further tests of the criteria and of Algorithm 2

The three tests use only the existing measured spectra and the band-design simulation; the scripts (`eA_band_truncation.py`, `eB_noise_sweep.py`, `eC_algorithm2_retro.py`) are included in the code repository.

S3.1 Band truncation of the measured spectra (Sect. 3.6)

Table S50 reports all 20 spectra. The thinned variant (five frequencies over 10–100 kHz) is shown for completeness but not interpreted: with five frequencies the Cole+L fit leaves only $2N - 5 = 5$ residual degrees of freedom, so both the estimate of σ and the nominal profile intervals are unreliable (they appear narrower than on the full band). On the remaining sub-bands, the predicted detection limit and the observed upper 95% bound of f_c were rank-correlated (Spearman 0.94 in the 12 KK-consistent spectra, 0.96 in all 20), also within spectra for changes relative to the full band (0.88 and 0.71). Restricted to sub-bands inside the domain of the first-order criterion ($\Gamma \geq 2^{1/\alpha}$; 38 of 60 KK-consistent cases), the correlation was 0.92 and the identifiability classification agreed with profile likelihood in 95% of

cases. Removing the three lowest instead of the three highest frequencies raised the detection limit by a median of 1.50 instead of -0.12 decade and the observed upper bound of f_c by 0.93 instead of -0.15 decade (12 of 12 KK-consistent spectra; exact Wilcoxon $p = 0.0005$; all 20 spectra: $p < 0.001$). One upper bound (all spectra, 40–100 kHz) reached the parameter bound.

Table S50: Band truncation of the measured spectra (all 20 spectra; nominal χ^2 threshold). df: residual degrees of freedom of the Cole+L fit ($2N-5$). Γ, detection limit $f_{\min}/\Gamma$ and observed upper 95% bound of f_c: medians. τ: number of spectra with practically identifiable $\log_{10}\tau$ (observed / predicted by Eq. (8)). Q, α: number identifiable (observed). Widths: median nominal profile widths of $\log_{10} Q$ (decades) and α.

Sub-band	kHz	N	df	Γ	$f_{\min}/\Gamma$ (kHz)	f_c upper (kHz)	τ obs/pred	Q/α ident.	Width Q/α
Full band	10–100	10	15	4.88	2.06	2.63	1 / 2	16 / 15	0.47 / 0.099
Lowest removed	20–100	9	13	1.77	11.36	5.53	0 / 0	13 / 13	0.80 / 0.166
Two lowest removed	30–100	8	11	0.99	30.66	12.42	0 / 0	9 / 9	1.17 / 0.239
Three lowest removed	40–100	7	9	0.45	95.54	17.18	0 / 0	7 / 7	1.59 / 0.324
Three highest removed	10–70	7	9	6.54	1.54	1.94	2 / 3	15 / 14	0.39 / 0.084
Thinned	10–100	5	5	8.52	1.22	1.98	6 / 5	16 / 16	0.35 / 0.074

S3.2 Noise sweep (Sect. 3.6)

The band-design simulation (log-spaced, $N = 20$, $f_{\max} = 100$ kHz; 60 truths) was repeated with the same noise draws scaled to 0.5, 1, 2.5 and 5% relative noise. The 2.5% level reproduced the original band-design runs exactly (identical identifiability classification and interval widths). Table S51 gives the identifiability of $\log_{10}\tau$. The closed-form classification agreed with profile likelihood in 97.5, 92.5, 91.3 and 87.5% of cases at the four noise levels. For profiles closed at three or more noise levels, the slope of $\log W_{\text{profile}}$ on $\log\sigma$ was 1.34 (interquartile range 1.21–1.49; 1.25 for $r = 0.3$–1, 1.36 for dispersions within the band, 0.54 in the few cases with $r < 0.3$), against 1 predicted by Eq. (8). A first attempt to test Eq. (9) by logistic regression on $\alpha\ln(f_{\min}/f_c)$, $\ln\sigma$ and $\ln(\alpha D)$ was uninformative because the ratio and D are strongly collinear in this design (bootstrap interval of the coefficient ratio -60 to 16). Regressing instead on $\ln\sigma$ and the combined geometry term $\alpha\ln(f_{\min}/f_c) - \ln(\alpha D)$, whose coefficients Eq. (9) predicts to be equal, gave a ratio of 1.43 (95% cluster-bootstrap interval 1.28–1.63): identifiability depends on noise more strongly than the first-order criterion implies.

Table S51: Noise sweep: identifiability of $\log_{10}\tau$ (% of 60 truths) by lower band limit and relative noise level (log-spaced, $N = 20$, $f_{\max} = 100$ kHz), and agreement of the closed-form classification with profile likelihood.

$f_{\min}$	0.5%	1%	2.5%	5%
10 kHz	68	60	47	30
3 kHz	92	80	70	58
1 kHz	98	93	85	75
0.3 kHz	100	100	97	93
Agreement of Eq. (8) (%)	97.5	92.5	91.3	87.5

Comparison with representative previous studies

Table S52 summarizes, in more detail than Table 1 of the main text, how the approach (Sect. 4.2) differs from representative previous studies. References are those of the main text.

Table S52: Representative approaches in selected previous studies and the approach taken in the main text.

Aspect	Representative approaches in selected previous studies	This study
Fitting-method evaluation	Comparison by residual error (Ayllón et al. 2009; Zhao et al. 2022; Zou et al. 2025)	Success at equal budget and wall-clock time against gradient and VarPro baselines; mechanism control
Linear parameters	Searched by population-based metaheuristics together with the nonlinear ones (Zhao et al. 2022; Zou et al. 2025); eliminated by VarPro in other settings (Sima and Van Huffel 2007; Dima et al. 2021; Simić et al. 2023)	Eliminated exactly by bound-constrained VarPro (Eq. (6)) on in vivo spectra
Structural identifiability	Structural identifiability and distinguishability of electrochemical models and circuits for noise-free data (Berthier et al. 1996, 2001; Alavi et al. 2017; Grandjean et al. 2017; Aghdam et al. 2022)	Practical identifiability for a finite band and noise: closed-form scaling of the width with $\sigma(f_{\min}/f_c)^{\alpha}/(\alpha D)$, bridging structurally identifiable and practically unidentifiable parameters
Practical identifiability	Low residual shown not to establish identifiability; injectivity, CRLB, profile likelihood and sensitivity analysis for battery circuits (Santoni et al. 2024); geometric identifiability under FIM-optimal sampling (Masoudi et al. 2026); profile likelihood (Raue et al. 2009)	Conditionally calibrated profile likelihood, checked against repeated sweeps
Frequency band	Numerical design of frequencies for simple RC circuits (Sebastià Bargues et al. 2022); CRLB-based frequency adjustment for a ten-parameter battery ECM (Sovljanski and Paolone 2024); closed-loop FIM-optimal EIS (Masoudi et al. 2026); Jacobian-sensitivity frequency selection for a simplified Cole circuit (Ouni et al. 2025)	Closed-form width, detection limit, effective-error limit and bias (Eqs. (8)–(11)); confounding angle ϑ explaining why a large sensitivity does not ensure identifiability; tested on measured sub-bands, across noise levels and retrospectively for Algorithm 2
Measurement chain	Time-delay compensation (Buendia et al. 2010); review of parasitic effects (Marcôndes et al. 2022)	Five corrections compared; effect on each descriptor quantified

Table S53: Complete benchmark: success rate (%) of the original (upper block) and variable-projection (VP, lower block) algorithms at 500 and 2,000 NFE (20 spectra $\times$ 25 runs). Success: NRMSE within 0.1% of the best-known value. L-SHADE-Restart and SAHO-LSHADE are reported in Table S3. Compact version: Table 9 of the main text.

Algorithm	M11 (4 par.)		M10 (5 par.)		M12 (8 par.)	
	500	2,000	500	2,000	500	2,000
HO	0.0	0.0	0.0	0.0	0.0	0.0
L-SHADE	0.0	79.6	0.0	19.2	0.0	1.2
ASHO-LSHADE	0.0	78.0	0.0	18.4	0.0	1.0
MS-TRF	100.0	100.0	100.0	100.0	20.4	66.0
L-SHADE+TRF	0.0	100.0	0.0	100.0	0.0	37.2
ASHO-LSHADE+TRF	0.0	100.0	0.0	100.0	0.0	40.2
VP-L-SHADE	100.0	100.0	95.8	100.0	13.6	80.2
VP-ASHO-LSHADE	100.0	100.0	95.8	100.0	13.6	80.8
VP-MS-TRF	100.0	100.0	100.0	100.0	81.4	97.8
VP-L-SHADE+TRF	100.0	100.0	97.8	100.0	16.0	97.0
VP-Screen+TRF	100.0	100.0	100.0	100.0	85.4	98.2

Table S54: Scope of validity of the main conclusions (moved from the main text, Sect. 4.5).

Conclusion	Evidence	Domain of validity	Data-specific?
VarPro: same minimum, fewer dimensions	Mathematical (Eq. (6))	Separable circuits	No
VarPro: more reliable search	Benchmark; mechanism control; per animal; leave-one-out	Separable circuits; small gain if weakly separable	Gain size
Nominal intervals under-cover; bootstrap corrects (conditionally)	6,000 simulated spectra; two error models; perturbed correlation	Under the estimated error structure; not under stronger correlation (83–88%)	Numbers
Closed form as scaling law and descriptor bias	Derivation; simulation; measured sub-bands; (f_c 0.1–30 kHz, α 0.60–0.95, $f_{\min}$ 0.3–10 kHz)	Trends and mechanism; as an interval anti-conservative (coverage 58–72%); $r \leq 0.5$; not with strong polarization	No
Operational detection limit f_{DL}	Monte Carlo coverage; classification study (2,160 cases); Algorithm 2 test; battery-type transfer (144 spectra)	Conservative (no false positives in domain); for a measured spectrum, FIM at the full fit or profile likelihood classify better; other circuits structurally, synthetic only	No
Band limit, not N, decides identifiability; band extension beats noise reduction	Simulation (incl. noise sweep); measured sub-bands with equal N	Only $N = 5$–20, log and linear grids, the evaluated bands and error models; not with strong polarization	Partly (sub-bands of one decade)
Algorithm 2 screening	Simulation test with priors exact to ± 1 decade and without prior	Balanced accuracy 0.83–0.89; not tested prospectively or with electrode polarization	Simulation-specific
Verdict for 10–100 kHz	20 spectra, 12 tissues, per animal	This band and instrument	Yes
Reproducibility limits	51 sweeps; 35 recordings	This instrument and protocol	Yes

Table S55: Numerical verification of the closed-form criterion (Appendix A, Step 9, of the main text; Note S4.13) on a representative spectrum (instrument band, $\sigma = 0.018$). $W_{(8)}$: Eq. (8); W_{lin}: exact linearized width; W_{prof}: expected-profile width (open: reaches the search bound); δD: relative error of the first-order band factor; bound: Eq. (A5). Widths in decades

r	$W_{(8)}$	W_{lin}	W_{prof}	δD	Bound
0.05	3.57	4.26	open	0.21	0.46
0.10	1.78	2.37	open	0.40	0.94
0.20	0.87	1.38	3.03	0.70	1.98
0.30	0.57	1.04	1.43	0.85	3.11
0.40	0.42	0.85	1.08	0.82	4.33
0.50	0.33	0.73	0.90	0.75	5.66
0.70	0.23	0.58	0.75	0.69	8.64
0.90	0.17	0.48	0.60	0.71	12.0

Table S56: Sandwich factor (Appendix A, Step 8, of the main text) κ_S, Eq. (A7), full-model sandwich factor κ_{full} (all five parameters) and ratio of the GLS width, Eq. (A6), to Eq. (8) on the instrument band: representative spectrum at $r = 0.3$ and range over the 16 quartile combinations of R_∞, Q, α, L (Table S12) and $r \in \{0.1, 0.3, 0.5\}$. Error models of Table S46 and Sect. 2.6 of the main text

Error model	κ_S	κ_{full}	GLS ratio
Estimated (51 sweeps)	1.34 (1.20–1.37)	1.33 (1.24–1.37)	1.23 (1.04–1.26)
$\rho_f + 0.2$	1.47 (1.29–1.52)	1.48 (1.34–1.53)	1.31 (1.09–1.35)
AR(1), $\rho = 0.6$	1.33 (1.31–1.36)	1.33 (1.28–1.36)	1.18 (1.15–1.21)
KK-only (27 sweeps)	0.70 (0.69–0.81)	0.80 (0.71–0.97)	0.66 (0.65–0.70)

Table S57: Transfer test of the closed-form criterion on a synthetic battery-type circuit $R_0 + \mathrm{ZARC} + A_W/\sqrt{j\omega} + j\omega L$ ($R_0 = 20$ mΩ; ZARC with $R = 15$ mΩ, $\alpha = 0.80$, $f_c = 20$ Hz; $A_W = 2$ mΩ s$^{-1/2}$; $L = 0.5$ μH), band $f_{\min}$–10 kHz with 10 log-spaced frequencies per decade. The criterion used the Jacobian of the remaining circuit (R_0, Q, α, A_W, L). $W_{(8)}$: Eq. (8); W_{lin}: exact linearized width; W_{prof}: expected-profile width (α re-optimized, linear coefficients by non-negative least squares). Identifiable ($W < 1$ decade): Eq. (8) / linearized / profile. Rows with $r > 0.5$ lie outside the domain of the criterion (dispersion in or near the band). Widths in decades; detection limit $f_{\min}/\Gamma$ in Hz. Script `ee_relevance.py`.

$f_{\min}$ (Hz)	N	σ (%)	r	$\sin\vartheta$	$W_{(8)}$	W_{lin}	W_{prof}	Identifiable	$f_{\min}/\Gamma$
200	18	0.2	0.16	0.13	1.33	2.35	1.20	no / no / no	28.5
200	18	0.5	0.16	0.13	3.32	5.87	open	no / no / no	89.6
100	21	0.2	0.28	0.15	0.38	0.85	0.43	yes / yes / yes	6.1
100	21	0.5	0.28	0.15	0.96	2.13	0.95	yes / no / yes	19.1
50	24	0.2	0.48	0.18	0.12	0.42	0.20	yes / yes / yes	1.4
50	24	0.5	0.48	0.18	0.30	1.05	0.43	yes / no / yes	4.4
20	28	0.2	1.00	0.21	0.03	0.21	0.10	yes / yes / yes	0.2
20	28	0.5	1.00	0.21	0.07	0.51	0.22	yes / yes / yes	0.7
10	31	0.2	1.74	0.23	0.01	0.07	0.02	yes / yes / yes	0.1
10	31	0.5	1.74	0.23	0.02	0.17	0.10	yes / yes / yes	0.2

Table S58: Stepped-sine sweep time from $f_{\min}$ to 100 kHz (log-spaced frequencies; integer cycles per frequency, settling and processing time excluded) and share of the time spent at the lowest frequency. Used in Fig. 6b of the main text, where the time needed to reduce white noise by averaging is scaled as $(\sigma_0/\sigma)^2$.

$f_{\min}$ (kHz)	N	Cycles	Sweep time (ms)	Lowest frequency (%)
10	10	5	2.0	24
10	10	10	4.1	24
10	20	5	4.0	13
10	20	10	8.0	13
3	10	5	5.1	33
3	10	10	10.1	33
3	20	5	9.6	17
3	20	10	19.3	17
1	10	5	12.4	40
1	10	10	24.8	40
1	20	5	23.0	22
1	20	10	46.1	22
0.3	10	5	35.0	48
0.3	10	10	70.0	48
0.3	20	5	63.1	26
0.3	20	10	126.3	26

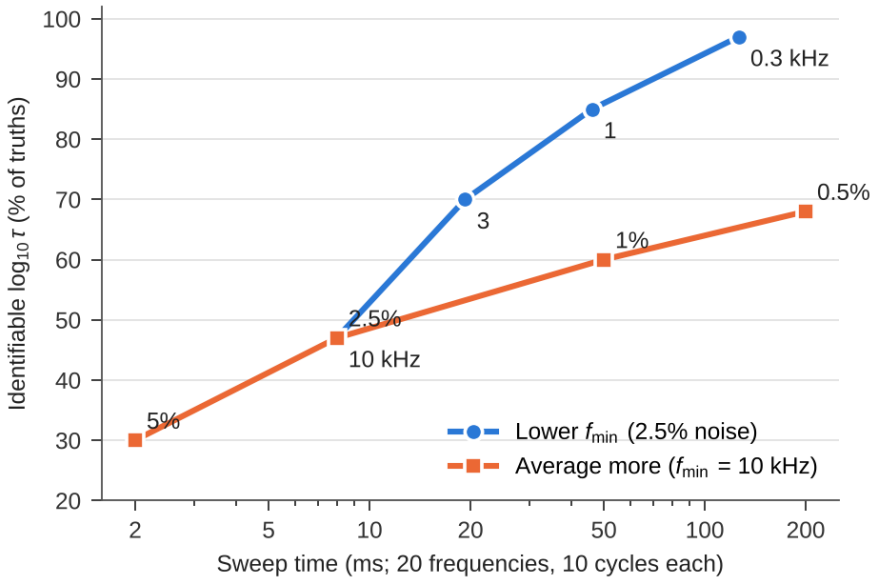


Figure S17: Identifiability of $\log_{10}\tau$ (noise sweep, 60 truths; Table S51) versus stepped-sine sweep time (Table S58) when $f_{\min}$ is lowered at 2.5% noise and when the noise is reduced by averaging at $f_{\min} = 10$ kHz, with averaging time scaled as $(\sigma_0/\sigma)^2$ (white noise; favourable to averaging).

Table S59: Monte Carlo check of the asymptotic widths (Appendix A, Steps 9 and 10, of the main text; Note S4.13): 300 noisy replicates per row of a synthetic Cole+L spectrum (median KK-consistent asymptote, $\alpha = 0.69$; instrument band, $N = 10$; independent relative Gaussian errors), fitted by variable projection over $(\log_{10}\tau, \alpha)$ with a global grid start. $W_{(8)}$: Eq. (8); W_{lin}: exact linearized width, Eq. (A8) (identical to the Wald width from the full Fisher information to 10^{-12}); empirical: $2z$ times the robust (MAD) and the ordinary standard deviation of $\log_{10}\hat{\tau}$; coverage (%) of $\hat{\tau} \pm W/2$ for the true value. Widths in decades. Script `mc_rigor.py`.

		Predicted		Empirical		Coverage (%)	
r	σ (%)	$W_{(8)}$	W_{lin}	$2z$MAD	$2z$SD	Eq. (8)	Eq. (A8)
0.3	1.8	0.57	1.04	0.97	1.33	72	91
0.5	1.8	0.33	0.73	0.68	7.53	69	91
0.5	0.8	0.15	0.32	0.30	0.31	70	94
0.7	0.8	0.10	0.26	0.25	0.28	58	92

Table S60: Plug-in detection limit $f_{\min}/\hat{\Gamma}$ computed, as in Algorithm 1, from the R–CPE–L fit of a single spectrum ($\hat{Q}$, $\hat{\alpha}$, $\hat{D}$) and $\hat{\sigma}$ from the Cole+L fit, against the value from the true asymptote; same circuit and band, $\sigma = 1.8\%$, 200 replicates per row. Noise-free plug-in: the bias that remains without noise, caused by the descriptor bias of Eq. (10) (fitted $\hat{\alpha}$ below the true 0.69). Frequencies in Hz; SD of $\log_{10}$ of the plug-in limit in decades. Script `mc_plugin.py`.

r	f_c	True	Noise-free plug-in	$\hat{\alpha}$	Median plug-in	5–95%	SD
0.02	34	829	865	0.678	829	539–1278	0.12
0.05	130	824	921	0.661	856	555–1295	0.12
0.10	355	816	1039	0.633	1034	579–1664	0.14
0.20	971	798	1408	0.577	1336	692–2083	0.15
0.30	1747	778	2123	0.524	1989	1025–3206	0.15
0.50	3662	737	7702	0.423	6908	3287–16059	0.22

Supplementary Note S4: Analyses added for the decision rule

S4.1 Number of bootstrap replicates

Table S61: Monte Carlo stability of the calibrated critical value c^* with the number of bootstrap replicates B (R–CPE–L truth with the median KK-consistent descriptors; instrument band; errors with the structure of Table S46 scaled to 0.018). Reference: $B = 10{,}000$. SD: standard deviation of c^* over repeated bootstrap runs; coverage: coverage of the interval obtained with c^* one SD below / above its mean. Script `boot_B.py`.

Parameter		$B = 200$	$B = 500$	$B = 1{,}000$	$B = 2{,}000$
$\log_{10} Q$ (ref. 6.85)	mean c^*	6.73	6.82	6.82	6.89
	relative SD (%)	12.1	8.6	5.4	3.5
	coverage (%)	93.0–96.0	93.7–95.8	94.2–95.6	94.6–95.4
α (ref. 7.27)	mean c^*	7.16	7.25	7.27	7.27
	relative SD (%)	11.4	8.1	4.9	3.2
	coverage (%)	93.1–96.0	93.6–95.8	94.2–95.6	94.5–95.4
$\log_{10} R_\infty$ (ref. 8.23)	mean c^*	8.11	8.15	8.19	8.24
	relative SD (%)	12.7	7.5	5.0	3.3
	coverage (%)	92.9–96.3	93.8–95.8	94.2–95.5	94.6–95.4

S4.2 Classification study with a Fisher-information baseline (Sect. 3.7)

Truths: the 60 band-design parameter sets of Sect. 2.8, re-simulated with an independent seed; bands $f_{\min} \in \{10, 3, 1, 0.3\}$ kHz to 100 kHz, $N = 20$ log-spaced; noise 1, 2.5 and 5%; three replicates (2,160 cases). Reference: profile-likelihood identifiability of $\log_{10} \tau$ (grid 0.05 decade on $[-7, 1]$, nominal threshold, closed and narrower than one decade) in each noisy spectrum. Plug-in rules use the R–CPE–L fit for the asymptote and $\hat{\sigma}$ from the Cole+L residuals ($2N - 5$ degrees of freedom), with $\hat{f}_c$ from the Cole+L fit. The FIM/CRLB baseline is the Wald width of $\log_{10} \tau$ from the Fisher information at the full Cole+L fit. Script `classify_sim.py`; metrics `analyse_class.py`.

Table S62: Classification of practical identifiability of $\log_{10}\tau$: full confusion matrices by subset and rule. Sens.: sensitivity; Spec.: specificity; BA: balanced accuracy.

Subset (n, % identifiable)	Rule	TP	TN	FP	FN	Acc.	Sens.	Spec.	BA
All (2160, 70%)	Majority (identifiable)	1509	0	651	0	0.70	1.00	0.00	0.50
	Closed form, plug-in	982	625	26	527	0.74	0.65	0.96	0.81
	Operational f_{DL}, plug-in	963	635	16	546	0.74	0.64	0.98	0.81
	FIM/CRLB, full fit	1497	571	80	12	0.96	0.99	0.88	0.93
	Closed form, true	1493	422	229	16	0.89	0.99	0.65	0.82
	W_{lin}, true	1459	541	110	50	0.93	0.97	0.83	0.90
In domain (306, 64%)	Majority (identifiable)	196	0	110	0	0.64	1.00	0.00	0.50
	Closed form, plug-in	195	86	24	1	0.92	0.99	0.78	0.89
	Operational f_{DL}, plug-in	107	110	0	89	0.71	0.55	1.00	0.77
	FIM/CRLB, full fit	196	95	15	0	0.95	1.00	0.86	0.93
	Closed form, true	192	63	47	4	0.83	0.98	0.57	0.78
	W_{lin}, true	183	92	18	13	0.90	0.93	0.84	0.89
Out of domain (1854, 71%)	Majority (identifiable)	1313	0	541	0	0.71	1.00	0.00	0.50
	Closed form, plug-in	787	539	2	526	0.72	0.60	1.00	0.80
	Operational f_{DL}, plug-in	856	525	16	457	0.74	0.65	0.97	0.81
	FIM/CRLB, full fit	1301	476	65	12	0.96	0.99	0.88	0.94
	Closed form, true	1301	359	182	12	0.90	0.99	0.66	0.83
	W_{lin}, true	1276	449	92	37	0.93	0.97	0.83	0.90
Near threshold (420, 44%)	Majority (identifiable)	184	0	236	0	0.44	1.00	0.00	0.50
	Closed form, plug-in	97	212	24	87	0.74	0.53	0.90	0.71
	Operational f_{DL}, plug-in	45	224	12	139	0.64	0.24	0.95	0.60
	FIM/CRLB, full fit	177	175	61	7	0.84	0.96	0.74	0.85
	Closed form, true	180	35	201	4	0.51	0.98	0.15	0.56
	W_{lin}, true	146	148	88	38	0.70	0.79	0.63	0.71
$\sigma = 1\%$ (720, 81%)	Majority (identifiable)	582	0	138	0	0.81	1.00	0.00	0.50
	Closed form, plug-in	517	125	13	65	0.89	0.89	0.91	0.90
	Operational f_{DL}, plug-in	412	138	0	170	0.76	0.71	1.00	0.85
	FIM/CRLB, full fit	582	130	8	0	0.99	1.00	0.94	0.97
	Closed form, true	580	100	38	2	0.94	1.00	0.72	0.86
	W_{lin}, true	572	131	7	10	0.98	0.98	0.95	0.97
$\sigma = 2.5\%$ (720, 72%)	Majority (identifiable)	517	0	203	0	0.72	1.00	0.00	0.50
	Closed form, plug-in	330	192	11	187	0.72	0.64	0.95	0.79
	Operational f_{DL}, plug-in	299	202	1	218	0.70	0.58	1.00	0.79
	FIM/CRLB, full fit	513	180	23	4	0.96	0.99	0.89	0.94
	Closed form, true	508	147	56	9	0.91	0.98	0.72	0.85
	W_{lin}, true	502	162	41	15	0.92	0.97	0.80	0.88
$\sigma = 5\%$ (720, 57%)	Majority (identifiable)	410	0	310	0	0.57	1.00	0.00	0.50
	Closed form, plug-in	135	308	2	275	0.62	0.33	0.99	0.66
	Operational f_{DL}, plug-in	252	295	15	158	0.76	0.61	0.95	0.78
	FIM/CRLB, full fit	402	261	49	8	0.92	0.98	0.84	0.91
	Closed form, true	405	175	135	5	0.81	0.99	0.56	0.78
	W_{lin}, true	385	248	62	25	0.88	0.94	0.80	0.87

End-to-end computing time per spectrum from the raw data (240 spectra, 2.5% noise; same machine, one thread, same starting grid on $\log_{10}\tau \in [-7, 1]$; script `timing_e2e.py`): R–CPE–L fit plus closed form 2.2 ms (interquartile range 1.3–2.4), R–CPE–L fit plus f_{DL} 2.2 ms (1.1–2.6), Cole+L fit (0.25-decade grid with local refinement) plus Wald width 37 ms (30–40), profile likelihood (161 points) 163 ms (130–190). For post-fit classification the plug-in rules also need $\hat{f}_c$ from the Cole+L fit and so cost about as much as the FIM baseline.

S4.3 Test of the revised Algorithm 2 (Sect. 3.7)

Pilot: 10–100 kHz, $N = 10$, 2.5% noise, one spectrum per truth (script `alg2_test.py`). The pilot identified τ in 20 of 60 truths (all with f_c of 10–30 kHz); for these, every candidate band was classified as adequate. For the others, the detection limit of each band was computed from the pilot asymptote and $\hat{\sigma}$. Wrong priors: $f_c^{\mathrm{exp}} = f_c \cdot 10^{\pm\delta}$ with a random sign per truth. Reference: majority of the three profile-likelihood replicates at 2.5% noise in the band.

Table S63: Revised Algorithm 2: confusion matrices for 240 band choices (73% identifiable; majority rule: accuracy 0.73, balanced accuracy 0.50) and for the 160 choices not settled by the pilot (61% identifiable). Decision: $f_c^{\mathrm{exp}} > f_{\mathrm{DL}}$ (operational) or $f_c^{\mathrm{exp}} > f_{\min}/\Gamma$ (closed form). Without the domain check added later (pilot exponent at a bound: refer); with it, 36 choices are referred and the decided ones reach balanced accuracies of 0.89, 0.88 and 0.86 (Note S4.14).

Cases	Prior	Rule	TP	TN	FP	FN	Acc.	Sens.	Spec.	BA
All 240	exact	operational	150	60	5	25	0.88	0.86	0.92	0.89
		closed form	160	55	10	15	0.90	0.91	0.85	0.88
	±0.5 decade	operational	143	59	6	32	0.84	0.82	0.91	0.86
		closed form	156	54	11	19	0.88	0.89	0.83	0.86
	±1 decade	operational	137	57	8	38	0.81	0.78	0.88	0.83
		closed form	147	51	14	28	0.82	0.84	0.79	0.81
Not settled by pilot (160)	exact	operational	72	60	3	25	0.82	0.74	0.95	0.85
		closed form	82	55	8	15	0.86	0.84	0.87	0.86
	±0.5 decade	operational	65	59	4	32	0.78	0.67	0.94	0.80
		closed form	78	54	9	19	0.82	0.80	0.86	0.83
	±1 decade	operational	59	57	6	38	0.72	0.61	0.91	0.76
		closed form	69	51	12	28	0.75	0.71	0.81	0.76

Without a prior, the median f_{DL} was above 32 kHz, 9.5 kHz (interquartile range 2.8–9.5), 1.1 kHz (0.18–3.2) and 76 Hz (18–406) for $f_{\min}$ of 10, 3, 1 and 0.3 kHz; values at the upper end of the search range ($3.16 f_{\min}$) mean that no dispersion along the pilot asymptote below that frequency is resolvable.

S4.4 Expanded battery-circuit transfer test (Sect. 3.7)

Table S64: Expanded synthetic transfer test: $R_0 + \mathrm{ZARC} + A_W/\sqrt{j\omega} + j\omega L$ ($R_0 = 20$ mΩ, $R_{\mathrm{ZARC}} = 15$ mΩ, $L = 0.5$ μH) over $f_c \in \{5, 20, 80\}$ Hz, $\alpha \in \{0.6, 0.8, 0.95\}$, $A_W \in \{10^{-6}, 2\times10^{-3}\}$ $\Omega\,\mathrm{s}^{-1/2}$, $f_{\min} \in \{10, 30, 100, 300\}$ Hz to 10 kHz (10 points per decade) and $\sigma \in \{0.2, 0.5\}$% (144 spectra). Reference: expected (noise-free) profile likelihood, identifiable if closed and narrower than one decade. Script `battery_grid.py`.

Subset (n, % identifiable)	Rule	TP	TN	FP	FN	Acc.	Sens.	Spec.	BA
All (144, 65%)	Closed form, Eq. (8)	94	46	4	0	0.97	1.00	0.92	0.96
	W_{lin}	76	50	0	18	0.88	0.81	1.00	0.90
$r \le 0.5$ (72, 31%)	Closed form, Eq. (8)	22	46	4	0	0.94	1.00	0.92	0.96
	W_{lin}	10	50	0	12	0.83	0.45	1.00	0.73
$r > 0.5$ (72, 100%)	Closed form, Eq. (8)	72	0	0	0	1.00	1.00	–	1.00
	W_{lin}	66	0	0	6	0.92	0.92	–	0.92

All 18 W_{lin} errors were false negatives. In these cases the nonnegativity bound of the Warburg or inductance coefficient was active in the reference profile on one side of the minimum, which truncates the constrained profile; the unconstrained linearization then overstates the width (e.g. $f_c = 80$ Hz, $\alpha = 0.8$, $f_{\min} = 300$ Hz, $\sigma = 0.2$%: $W_{\mathrm{lin}} = 1.08$, profile 0.40 decade). The four closed-form errors were false positives with profile widths of 1.0–1.17 decades.

S4.5 Three-way rule (Sect. 3.7)

Table S65: Three-way rule: *likely* if both the closed form ($f_{\min}/\Gamma$) and the operational limit (f_{DL}) predict identifiability, *unlikely* if neither does, *uncertain (refer)* otherwise. Decided: cases classified as likely or unlikely; errors and metrics refer to decided cases. Algorithm 2: the 160 band choices not settled by the pilot. Scripts `threeway.py`, `analyse_sens.py`.

Setting	n	Decided	FP	FN	Referred	% identifiable among referred	Sens.	Spec.
Classification, in domain	306	194	0	1	112	79	0.99	1.00
Classification, near threshold	420	294	0	68	126	71	0.28	1.00
Classification, all	2160	1635	0	295	525	92	0.71	1.00
Algorithm 2, exact prior	160	137	3	11	23	78	0.86	0.95
Algorithm 2, ±0.5 decade	160	134	3	16	26	73	0.79	0.95
Algorithm 2, ±1 decade	160	140	5	27	20	60	0.68	0.91
Battery-type circuit, $r \le 0.5$	72	56	0	0	16	75	1.00	1.00
Battery-type circuit, all	144	122	0	0	22	82	1.00	1.00

S4.6 Algorithm 2 when the candidate band differs from the pilot (Sect. 3.7)

Pilot as in S4.3 (10–100 kHz, $N = 10$, 2.5% white noise). Reference: profile-likelihood identifiability in the candidate band (majority of three replicates) under the stated condition. White-noise mismatch uses the reference spectra of the classification study at 1% and 5%; the other conditions were simulated anew (script `alg2_sens.py`). Electrode CPE: series $K/(j\omega)^{0.6}$ with $|Z_{\mathrm{pol}}|$ equal to 10% or 50% of the tissue impedance at 1 kHz, present in pilot and candidate band, not in the fitted model.

Table S66: Algorithm 2 (two-way decision $f_c^{\mathrm{exp}} > f_{\mathrm{DL}}$; pilot-identified truths settled as adequate) under conditions in the candidate band that differ from the pilot; 240 band choices each. Three-way: decided / referred / errors among the choices not settled by the pilot.

Condition in candidate band	Prior	% ident.	FP	FN	Sens.	Spec.	BA	Three-way
Same as pilot (2.5%)	exact	73	5	25	0.86	0.92	0.89	137 / 23 / 14
	±0.5 dec.	73	6	32	0.82	0.91	0.86	134 / 26 / 19
White noise 1%	exact	80	2	39	0.80	0.96	0.88	137 / 23 / 21
	±0.5 dec.	80	2	45	0.77	0.96	0.86	134 / 26 / 28
White noise 5%	exact	56	27	7	0.95	0.74	0.85	137 / 23 / 16
	±0.5 dec.	56	26	12	0.91	0.75	0.83	134 / 26 / 17
$\sigma = 2.5\%\,(10\ \mathrm{kHz}/f)^{0.3}$ below 10 kHz	exact	68	17	22	0.86	0.78	0.82	120 / 20 / 18
	±0.5 dec.	68	16	21	0.87	0.79	0.83	124 / 16 / 20
Electrode CPE, 10%	exact	69	3	23	0.86	0.96	0.91	143 / 25 / 12
	±0.5 dec.	69	4	28	0.83	0.95	0.89	146 / 22 / 19
Electrode CPE, 50%	exact	38	79	11	0.88	0.47	0.68	159 / 25 / 59
	±0.5 dec.	38	77	14	0.84	0.49	0.67	160 / 24 / 61

S4.7 Reproduction from the public data

The pipeline of this repository (`pipeline/`) was run on version 2 of the public dataset (Zenodo record 20408128, `PreparedData`, subsets A and D; converter `prepare_falagario.py`; $B = 200$). It found the same 20 complete spectra and the same 51 complete sweeps, and it reproduced the error model of Table S46 from the sweeps ($\sigma_0 = 0.110$, $\gamma = 0.009$, $\rho_f = 0.180$, $\rho_{\mathrm{RI}} = -0.194$, $\kappa = 0.459$). For 11 of the 20 spectra (1, 4, 7, 8, 13–15, 17–20) the median spectrum was identical, and the fit, KK, error and criterion quantities agreed with the published values within their rounding; the upper bound of f_c differed by at most 0.02 decade, within the resolution of the profile grid, and $\hat{f}_c$ of the bladder, whose τ profile is flat, by 0.24 decade (Table S67). For the other nine (2, 3, 5, 6, 9–12, 16), the public median spectra fitted better than those analysed here (M11 NRMSE lower by 10–40%), while their complete sweeps were identical; the public version therefore differs from the one analysed here. The raw files of the record (`Supplementary.zip`, `OriginalData`), converted with the gain and system phase of the dataset's `Data_Freq_Processing.m` with or without its moving-median outlier removal (`raw_to_impedance.py`), reproduced the same 11 spectra but not the other nine either. All nine come from recordings with two or more sweeps, in which the per-frequency medians of the real and imaginary parts can switch between drifting sweeps: removing 1–10 of the 360–1,180 readings of such a recording changed its M11 NRMSE by up to a factor of two (e.g. tendon 0.025–0.046, published 0.036; muscle 0.088–0.227, published 0.201), whereas recordings with one sweep or stable sweeps did not change. The published values of the nine spectra are thus consistent with a data version that differs from version 2 by a few readings, which is no longer available. On the public data these nine changed no identifiability verdict and no closed-form prediction, but three of them (3, 6, 11) became KK-consistent. For multi-sweep recordings, a spectrum built from the median of the per-sweep spectra would be less sensitive to individual readings.

Table S67: Reproduction of the published per-spectrum results from the public data with the pipeline of this repository. Upper block: the 11 spectra with identical median spectra; number agreeing with the published value within its rounding or within 0.5%. Lower block: classifications for all 20 spectra.

Quantity (11 spectra with identical data)	Within rounding or 0.5%
M11 NRMSE	11/11
KK RMS residual (%)	11/11
KK maximum residual (%)	11/11
σ	11/11
α (M11)	11/11
Γ	11/11
Closed-form detection limit (Hz)	10/10
$\hat{f}_c$ (Hz)	10/11; bladder 10 vs 6 (flat profile)
Upper bound of f_c (Hz)	3/11; all within 4.4% (0.02 decade)
M11 NRMSE	0.2%
KK RMS residual	0.3%
KK maximum residual	0.2%
σ	2.4%
α (M11)	0.8%
Γ	inf%
$\hat{f}_c$	73.1%
Upper bound of f_c	4.4%
Classification (all 20 spectra)	Agreement
Practical identifiability of $\log_{10}\tau$ (observed)	20/20
Closed-form prediction	20/20
KK consistency (3% threshold)	17/20

S4.8 Sensitivity analysis: public data with spectra from the median of the per-sweep spectra

The construction of the spectra used in this study (per-frequency medians of all readings, real and imaginary parts separately) is sensitive to a few readings in recordings whose sweeps drift (Note S4.7). As a sensitivity analysis, the measured-data analysis was repeated on version 2 of the public data with each spectrum built as the per-frequency median of the per-sweep median spectra of its complete sweeps (pipeline option `-spectrum-method sweep_median`, the default of this repository; $B = 200$; results in `pipeline/results_v2_sweepmedian/`). With this construction, removing 1–10 readings left the M11 fit of 18 of the 20 spectra unchanged (largest change otherwise a factor of 3, in a recording with a single complete sweep), against changes of up to a factor of 2.4 for the pooled median. The error model of Table S46 was reproduced exactly. Table S68 compares the main measured-data results.

Table S68: Main measured-data results with the spectra of this study (pooled medians) and with spectra from the median of the per-sweep spectra on the public data (version 2). Identifiability with the nominal threshold unless stated; KK: Kramers–Kronig.

Result	This study	Per-sweep median, public data
KK-consistent spectra (3% threshold)	12	17
KK-inconsistent spectra	8	3 (muscle, peritoneum, liver)
M11 preferred by AICc among KK-consistent	10 of 12	14 of 17
$\log_{10}\tau$ identifiable, nominal / calibrated	1 / 0	2 / 0
$\log_{10}\Delta R$ identifiable, nominal / calibrated	2 / 1	3 / 2
$\log_{10} Q$, α, $\log_{10} R_\infty$ identifiable in KK-consistent spectra	12 / 12 / 12	17 / 17 / 17
Median calibrated width of $\log_{10} Q$ / α (decade / –)	0.62 / 0.137	0.50 / 0.107
Median Γ (20 spectra)	4.9	8.6
Closed-form detection limit, median of KK-consistent (kHz)	1.1	1.1
Operational f_{DL}, median of KK-consistent (kHz)	–	2.8
Closed-form prediction of τ identifiability	19 of 20	19 of 20
Three-way rule (decided cases correct; referred)	–	17 of 17; 3
Band truncation: Q, α identifiable on 40–100 kHz	7 of 12	11–12 of 17

The conclusions of the main text hold: the relaxation time remains non-identifiable with the calibrated threshold in every spectrum, the R–CPE–L descriptors remain identifiable in every KK-consistent spectrum, and the detection limit of the 10–100 kHz band is unchanged. Two results change. Five of the eight spectra classified here as KK-inconsistent are KK-consistent when built from per-sweep spectra, so most of the KK inconsistency reported in Sect. 3.1 reflects drift between sweeps mixed by the pooled median rather than inconsistency within a sweep. And one further muscle spectrum has an identifiable τ with the nominal threshold (width 0.73 decade; $\hat{f}_c = 2.0$ kHz), which the closed form predicts and the three-way rule refers as uncertain.

S4.9 Drift between sweeps and its effect on the Kramers–Kronig test

The raw files of the public record (`Supplementary.zip`) carry the duration of each reading, which gives the elapsed time of every sweep. They were converted as in Note S4.7 (`raw_to_impedance.py` with outlier removal) and analysed with `drift_analysis.py`. A sweep of 10 frequencies with about 20 readings each took a median of 9.5 s in the first animal and 158 s in the second; consecutive sweeps followed about 9.8 s apart. Between consecutive sweeps, the largest change of $|Z|$ over the band was a median of 1.1% per second (interquartile range 0.6–3.2%; 0.11–10.6%) and of the phase 0.24° per second (14 recordings with two or more sweeps, 56 sweeps). Over a recording, $|Z|$ differed between sweeps by a median of 25% (1–112%). The KK residual of the pooled-median spectrum rose with this spread (Spearman 0.85, $p = 10^{-4}$) and with the drift rate (0.78), whereas the individual sweeps were KK-consistent in all but four recordings, and the median of the per-sweep spectra was KK-consistent in 11 of these 14 recordings (Table S69). Drift, or probe movement, between sweeps rather than inconsistency within a sweep therefore caused most of the KK violations of the pooled spectra.

Table S69: Between-sweep drift and KK residual (RMS, %) of the 14 recordings with two or more complete sweeps (raw public files). Drift: median over consecutive sweep pairs of the largest relative change of $|Z|$ over the band per second; spread: largest relative range of $|Z|$ across sweeps.

#	Tissue	Sweeps	Drift (%/s)	Spread (%)	KK pooled	KK per-sweep median	KK worst sweep
2	tendon	2	1.10	11	1.34	1.28	1.38
3	muscle	2	3.30	28	2.06	1.43	1.58
4	bladder	2	0.11	1	1.27	1.25	1.26
5	muscle	2	6.24	56	5.18	2.04	4.49
7	muscle	3	1.09	16	1.26	1.26	1.40
8	muscle	3	10.63	112	12.33	12.10	24.36
9	intestine	4	1.04	23	1.20	1.26	1.45
10	peritoneum	4	7.60	89	8.01	5.59	10.07
11	liver	4	0.77	21	1.80	1.29	1.44
12	liver	6	1.21	73	5.25	4.87	37.97
13	kidney	4	0.31	4	1.32	1.29	1.39
14	artery	5	0.18	4	1.33	1.33	1.36
15	spleen	5	0.55	27	1.41	1.41	1.87
16	stomach	4	2.77	58	3.01	1.40	1.57

S4.10 Dispersion above the band: two-sided identifiability window

Synthetic Cole+L spectra with tissue-like parameters (25 truths: R_∞ 500–5,000 Ω, ΔR one to three times R_∞, α 0.60–0.95, L 0.4–2.5 mH), f_c at 1.5, 3, 10 and 30 times $f_{\max}$, bands of two decades ($N = 20$, log-spaced) ending at 10, 30 and 100 kHz, relative noise 0.5% and 2.5% (600 cases). Reference: expected (noise-free) profile likelihood of $\log_{10}\tau$, closed and narrower than one decade, computed without bounds on the linear coefficients (identifiability from the data alone) and with the nonnegativity bounds used in the fits. Scripts `window_sim.py`, `analyse_window.py`, `window_constraint_check.py`.

Table S70: Dispersion above the band: classification of the identifiability of $\log_{10}\tau$ by the mirrored closed form, Eq. (12), and by the exact linearized width W_{lin}, against the unconstrained and the bounded expected profile likelihood; $r' = (f_c/f_{\max})^{-\alpha}$.

Reference	Cases	% identifiable	Rule	FP / FN	Accuracy	BA
Unconstrained	All (600)	12	Closed form	92 / 0	0.847	0.912
		12	W_{lin}	6 / 0	0.990	0.994
	$r' \le 0.5$	0	Closed form	63 / 0	0.858	0.429
		0	W_{lin}	6 / 0	0.986	0.493
	$r' > 0.5$	48	Closed form	29 / 0	0.814	0.821
		48	W_{lin}	0 / 0	1.000	1.000
With bounds	All (600)	39	Closed form	0 / 68	0.887	0.855
		39	W_{lin}	0 / 154	0.743	0.672
	$r' \le 0.5$	21	Closed form	0 / 29	0.935	0.842
		21	W_{lin}	0 / 86	0.806	0.533
	$r' > 0.5$	92	Closed form	0 / 39	0.750	0.864
		92	W_{lin}	0 / 68	0.564	0.762

Identifiable fraction by $f_c/f_{\max}$ = 1.5, 3, 10, 30: unconstrained 50%, 0%, 0%, 0%; with bounds 93%, 59%, 4%, 0%. Of the 235 cases identifiable with bounds, 160 were not identifiable from the data alone. Median $\sin\vartheta' = 0.08$; for $r' > 0.5$ and closed unconstrained profiles, W_{lin} was a median 1.01 times the profile width. A direct check (12 truths,

$f_{\max} = 100$ kHz, 0.5% noise) gave, at $f_c = 3f_{\max}$, unconstrained profile widths of 4.0–4.5 decades against bounded widths of 0.5–0.9 decade and W_{lin} of 1.0–1.6 decades.

S4.11 Prospective test on measured battery spectra

Data: Zhang et al. (2020; Zenodo 10.5281/zenodo.3633835, `EIS data.zip`, CC BY 4.0), commercial LCO/graphite coin cells, 60 frequencies from 20 kHz to 0.02 Hz. Subset: state of charge V, 25 °C, cells 1–8, four cycles per cell spread over the ageing record (32 spectra). Circuit R_0 + ZARC$_1$ + ZARC$_2$ + $A_W/\sqrt{j\omega} + j\omega L$ (NNLS for the linear coefficients; ZARC$_2$ restricted to $f_{c2} \geq 63$ Hz to prevent label switching; full-band NRMSE median 1.3%). The asymptote for the criteria is the one implied by the fit on each truncated band ($Q_1 = \tau_1^{\alpha_1}/\Delta R_1$), because a separate R–CPE fit is degenerate on truncated bands where the CPE and Warburg power laws are nearly collinear. Reference: profile likelihood of $\log_{10}\tau_1$ on the truncated band (nominal threshold), without bounds (data alone) and with the nonnegativity bounds of the fit; accuracy: deviation of the truncated-band estimate from the full-band τ_1. Scripts `battery_measured.py`, `analyse_battery_measured.py`.

Table S71: Measured battery spectra, 224 truncated bands ($f_{\min} = 1$–100 Hz, $f_{\max} = 20$ kHz): prediction of the identifiability of $\log_{10}\tau_1$ from each truncated band, against the unconstrained profile likelihood (identifiability from the data) and against accurate identification (identifiable and within 0.5 decade of the full-band value). Three-way: decided cases only.

Reference	Cases	Rule	n	% positive	FP / FN	Accuracy	BA
Data-only identifiability	All	Closed form	224	52	59 / 0	0.74	0.72
		f_{DL}	224	52	29 / 6	0.84	0.84
		Three-way	188	59	29 / 0	0.85	0.81
	Dispersion below band	Closed form	138	36	43 / 0	0.69	0.76
		f_{DL}	138	36	17 / 6	0.83	0.84
		Three-way	106	42	17 / 0	0.84	0.86
Accurate identification	All	Closed form	224	35	98 / 0	0.56	0.66
		f_{DL}	224	35	63 / 1	0.71	0.78
		Three-way	188	41	63 / 0	0.67	0.72
	Dispersion below band	Closed form	138	8	82 / 0	0.41	0.68
		f_{DL}	138	8	51 / 1	0.62	0.75
		Three-way	106	9	51 / 0	0.52	0.73

Full band: τ_1 identifiable in 32 of 32 spectra (f_{c1} 2.3–33 Hz, α_1 0.79–1.00, f_{c2} 64–549 Hz). Truncated bands, data-only identifications: within the band (86 cases) 67 identified, all within 0.5 decade of the full-band value (median deviation 0.07 decade); below the band (138 cases) 50 identified, 11 within 0.5 decade (median deviation 1.24 decades, interquartile range 0.60–1.50, always towards higher f_c). Median truncated-band estimate relative to the full band: 0.84, 0.83, 0.92, 3.17, 3.90, 4.27, 7.23 for $f_{\min}$ = 1, 3, 10, 20, 30, 50, 100 Hz. The operational rule and the three-way rule rated 34 of the 39 inaccurate identifications as likely. The fitted f_{c1} of 36 of these 39 lay inside the truncated band (median 4.8 times $f_{\min}$), so the requirement $f_{\min} \leq \hat{f}_c$ of the decision rule rejected only 3 of them; it kept all 78 accurate identifications.

S4.12 Summary: what each criterion achieves

Table S72 collects, for each objective, the validation, the result, the conditions of use and the limitations of the criteria; it was Table 8 of the main text in the previous version.

Table S72: (Formerly Table 8 of the main text.) What the closed-form and operational criteria achieve, separated by objective; reference: profile likelihood; $r = (f_{\min}/f_c)^{-\alpha}$

Objective	Validation	Result	Applicability	Limitations
Scaling law for $\log_{10}\tau$ (Eqs. (8)–(11) of the main text)	Band-design simulation; noise sweep; measured sub-bands; Monte Carlo	Correct trends in $f_{\min}/f_c$, α and σ; explains loss of τ by confounding ($\sin\vartheta$); sub-band detection limits rank-correlated with observed bounds (Spearman 0.94)	Dispersion below the band, $r \le 0.5$, no strong electrode polarization	Anti-conservative: $\widetilde{W}_{95} \approx W_{\mathrm{lin}}/2$, coverage 58–72%; 229 false positives of 651 at the true parameters; widths grew faster than proportionally with noise
Bias of the R–CPE–L descriptors (Eq. (10))	Numerical refits of noise-free Cole spectra (Table S25)	Within 2% for $f_{\min}/f_c \ge 33$, within 3–11% at 10 and 3.3	Dispersion below the band	Linearized; not a correction for misspecified intervals (coverage 10–22%)
Operational detection limit f_{DL} (W_{lin}, Eq. (A8))	Classification study (2,160 cases); Monte Carlo coverage	Coverage 91–94%; in domain no false positives; measured spectra: median f_{DL} 3.0 kHz; on the full band it removed the closed-form false positive but missed the one identifiable spectrum	Design of a band not yet measured; quadratic likelihood	Conservative: sensitivity 0.55 in domain, 0.24 near the threshold; 0.45 in the battery test, where bounds were active
Three-way rule (closed form and f_{DL} combined)	Same study; Algorithm 2 test; battery test	No false positives over 2,160 cases; in domain 193 of 194 decided cases correct	Screening; uncertain cases referred	Refers 24% of all and 37% of in-domain cases
Screen candidate bands (Algorithm 2)	60 truths × 4 bands; priors exact, ±0.5, ±1 decade, none; band conditions differing from the pilot	Balanced accuracy 0.89, 0.86 and 0.83 (majority 0.50); with the domain check, 0.89, 0.88 and 0.86 on the 85% of choices decided (Note S4.14); three-way 0.91 on 137 of 160 unsettled choices; without prior, f_{DL} per band	Pilot spectrum; error level of the candidate band known or bounded	Not tested prospectively; false positives rise from 5 to 17–27 with noise the pilot does not see and to 79 with an electrode CPE of 50%
Measured battery spectra	224 truncated bands, public data	BA 0.84 (f_{DL}), 0.72 (closed form); identified τ_1 accurate 67/67 within, 11/50 below the band	Extra nuisance elements	Model discrepancy undetected
Classify a spectrum after fitting	Same study; 20 measured spectra	FIM/CRLB at the full fit: balanced accuracy 0.93, 37 ms; profile likelihood 163 ms; measured: 19 of 20, equal to the majority rule	Any spectrum with a Cole+L fit	Closed-form rules not intended for this; one identifiable measured spectrum, so sensitivity is not estimable

S4.13 Numerical verification of Appendix A (former Step 9 and Monte Carlo part of Step 10)

Verification of Steps 1–8. On synthetic Cole+L spectra with the median asymptote of the 12 KK-consistent spectra ($r = 0.05$–0.9; Table S55), the remainder ratio of Step 1 never exceeded r, the bound of Eq. (A5) held in every case, the first-order width was 0.45–0.64 times the exact linearized width for $r = 0.3$–0.5, and all sandwich factors equalled one for independent errors of equal variance (`verify_criteria.py`). On this representative spectrum, $\sin\vartheta \approx 0.22$ on the 10–100 kHz band: only about 22% of the norm (5% of the energy) of the second-order signal lies outside what a change of R_∞, Q, α and L can absorb; lowering $f_{\min}$ from 10 to 0.3 kHz raised $\sin\vartheta$ from 0.17 to 0.54.

Exact linearized width. Eq. (A8) agreed with the Wald width from the full Fisher information to 10^{-12}. In a Monte Carlo check (300 replicates per condition; Table S59), $2z$ times the empirical (or, near the boundary, robust) standard deviation of $\log_{10}\hat{\tau}$ was within 3–8% of W_{lin}, and intervals $\hat{\tau} \pm W_{\mathrm{lin}}/2$ covered the truth in 91–94% of replicates, against 58–72% for the first-order width $\widetilde{W}_{95}$.

Plug-in detection limit. Computed from fitted rather than true descriptors, the detection limit is biased upwards (conservatively) as the dispersion approaches the band, by less than 5% for $r \le 0.05$ and by a factor of 2.7 at $r = 0.3$, with a scatter of 0.12–0.15 decade at $\sigma = 1.8\%$ (Table S60).

S4.14 Measured tetrapolar spectra

Data. Public compilation of Dell'Osa et al. (Zenodo, 10.5281/zenodo.15795175, CC BY 4.0), file registros_EIS.mat. Primary set: the 50 spectra labelled error-free (IDs 1–50), from Ayllón et al. (2016): human, tetrapolar, SFB7 (ImpediMed), 256 frequencies from 3.08 kHz to 1 MHz; the frequency column of this subset is angular frequency (rad/s) and

was converted to Hz. The 50 spectra fall into five recording series by their resistance at the lowest frequency (6, 22, 2, 10 and 10 spectra; two groups with f_c of 44 and 77 kHz), so the number of independent units is small; the same 50 spectra also appear in the compilation as error class S4. Every spectrum was thinned to the frequencies nearest to 10 log-spaced points per decade (27 frequencies). Secondary set: 36 spectra of a Xitron 4001B analyzer (tetrapolar, human, 5 kHz–1 MHz, 50 frequencies in kHz; reactance stored with positive sign and inverted here), from Gonzalez-Correa (2019; J. Phys.: Conf. Ser. 1272, 012018). All 1,180 spectra labelled with measurement errors (classes S2–S7) passed the linear KK test (RMS $\leq 0.97\%$), so KK consistency did not detect these errors.

Analysis (`pipeline/wideband_test.py`, options `-per-decade 10 -fmin-step 0.125`). Band truncation from below in steps of 0.125 decade with $f_{\max} = 1$ MHz fixed and at least one decade retained; on each band, Cole+L fit, profile likelihood of $\log_{10}\tau$ (nominal threshold, closed and narrower than one decade), closed form and operational rule from the R–CPE–L asymptote of that band, and the Wald width at the full Cole+L fit (Fisher baseline). An identified value is accurate if within 0.5 decade of the full-band value. Algorithm 2: pilot = top decade (0.1–1 MHz), f_{DL} and W_{lin} evaluated for each candidate band with priors exact, ± 0.5 and ± 1 decade.

Table S73: Measured tetrapolar spectra: band truncation and prediction of the identifiability of $\log_{10}\tau$ from each truncated band. BA: balanced accuracy; FP/FN: false positives/negatives.

Dataset	Rule	Cases	Identifiable	BA	Sens.	Spec.	FP / FN
SFB7, error-free (50 spectra)	Closed form	650	280	0.83	0.66	1.00	0 / 94
	Operational (W_{lin}, f_{DL})	650	280	0.50	0.00	1.00	0 / 280
	Three-way (464 decided)	464	94	0.50	0.00	1.00	0 / 94
	FIM at full fit	650	280	0.93	0.86	1.00	0 / 40
	Algorithm 2 (all priors)	650	280	0.50	0.00	1.00	0 / 280
Xitron (26 KK-consistent)	Closed form	286	3	0.64	0.33	0.94	17 / 2
	Three-way (268 decided)	268	2	0.50	0.00	1.00	0 / 2
	FIM at full fit	286	3	0.50	0.00	1.00	0 / 3

SFB7 set: full band, τ identifiable in 50 of 50 (95% width 0.17–0.24 decade), α 0.48–0.50, effective error 0.8–1.6%, KK RMS $\leq 0.17\%$. Largest $f_{\min}$ with identifiable τ: 7.8 kHz for the series with $f_c = 44$ kHz ($f_c/f_{\min} = 5.6$–5.8) and 24.8 kHz for $f_c = 77$ kHz (3.1–3.2); the closed form predicted 6.2 and 9.8 kHz. The fitted f_c fell as $f_{\min}$ rose (median ratio of identified values to the full-band value 0.72 at 6.2 kHz, 0.50 at 12.2 kHz, 0.17 at 24.8 kHz). Of 280 identified values, 240 were within 0.5 decade of the full-band value (median deviation 0.12 decade) and 40 (series with $f_c = 77$ kHz, $f_{\min} = 19.3$ and 24.8 kHz) were 0.65–0.80 decade too slow, each with $\hat{f}_c < f_{\min}$; the rule "identifiable and $\hat{f}_c \geq f_{\min}$" kept 240 of 240 accurate and rejected 40 of 40 inaccurate values. The three-way rule referred 186 cases, all identifiable. The R–CPE–L exponent was at its lower bound (0.3) on every band and in the pilot, because no band of one decade lay sufficiently above f_c ($r = (f_{\min}/f_c)^{-\alpha} \leq 0.5$ requires $f_{\min} \geq 4 f_c$ approximately at $\alpha = 0.5$, that is, 180–310 kHz); the asymptote, and hence W_{lin}, f_{DL} and Algorithm 2, were outside their domain and predicted no identifiability anywhere. The pilot (0.1–1 MHz, 11 frequencies) identified τ in none of the 50 spectra (95% profile width ≥ 3.3 decades) and its exponent was at the bound in all, so the domain check now part of Algorithm 2 (refer when the pilot exponent lies at a bound) classifies every candidate band as uncertain. Applied to the simulation of Table S63 (`alg2_domain.py`), the same check referred 36 of 240 band choices (15%; 72% of them identifiable), and the balanced accuracy of the decided choices was 0.89, 0.88 and 0.86 for priors exact, ± 0.5 and ± 1 decade (without the check: 0.89, 0.86 and 0.83).

Xitron set: 26 of 36 spectra KK-consistent; effective error 3.5–6.2%; fitted f_c at or above $f_{\max}$ in 20 (19 at the upper bound of τ) and α at its lower bound in 6; τ identifiable on the full band in 1 of 26, consistent with the identifiability window (Eq. 12). The closed form, which assumes a dispersion below the band, gave 17 false positives, 10 of them with $\hat{f}_c > f_{\max}$; the three-way rule gave none.